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Named Entity Recognition in German-language Telegram channels

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Abstract

This thesis investigates the potentials and limitations of Named Entity Recognition (NER) approaches for the automated analysis of German-language text data from the messenger service Telegram. I will focus on the standard named entity types person, location, and organization, and the non-standard entity types time, date, and action. The thematic focus is on the mobilization for protests against the COVID-19 measures in Germany.

My thesis provides an introduction to the task of NER and to prior work in the context of shared tasks for German-language and social media NER. It explores the applicability of two existing NER models provided by the spaCy library and the Flair framework. I furthermore extensively explore the potentials and limitations of building a custom solution for my concern by fine-tuning pretrained transformer models, including the task of creating an annotated Telegram corpus for NER including non-standard entity types. My work contributes to the development of German-language NER in user-generated texts from messenger services, exploring the potentials of state of the art approaches for this purpose.

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Chapter 1

Introduction

The outbreak of the COVID-19 pandemic was followed by an increasing spread of misinformation and conspiracy theories worldwide (cf. Friedrich Naumann Stiftung 2020). Those narratives evolved not only around the virus itself, but also about the measures for its containment, including the mandatory use of face masks, rapid testing, and vaccinations.

Online conspiracy narratives soon began to manifest themselves in protests on the street. The so-called 'Querdenken' movement became one of the most prominent network organizing protests against the COVID-19 containment measures. Its main actors were eager to present 'Querdenken' as a diverse, peaceful movement from the middle of society. However, extremist right-wing networks and conspiracy theorists were involved in the protests from the beginning on (cf. Forschungsinstitut Gesellschaftlicher Zusammenhalt 2020). The movement reached its preliminary climax with several large demonstrations taking place in Berlin. In the context of these demonstrations, violent clashes occurred between police and protesters, which often refused to act according the official hygiene restrictions such as wearing a face mask. Furthermore, in several incidents journalists were violently attacked by protesters and accused of being part of the 'Lügenpresse' (lying press). On the 29th of August 2020, protestors around five hundred protesters, among them several prominent actors from the German far-right movement, broke the barriers in front of the Reichstag in Berlin, and occupied the stairs of the building, some of them waving 'Reichskriegsflaggen' (cf. rbb24 2020).

On April 21st 2021, the last large authorized demonstration of the 'Querdenken' movement took place in Berlin. After police had decided to disperse the demonstration du to several violations of official restrictions by demonstration participants, protesters moved to areas nearby where clashes between protesters and police continued to occur throughout the rest of the day. The subsequent 'Querdenken' demonstrations were prohibited by authorities, but nonetheless several hundred protesters gathered in different spots in Berlin e.g. on the 21st, 28th and 29th of August 2021.

General restrictions for demonstrations and public gatherings in the context of the following lockdowns decreased the momentum of the large, centralized protests. However, the protests did not stop, but became more de-centralized and informal: In several places, people gathered for vigils or walks to protest against the government's COVID-19 politics (cf. CeMAS - Center für Monitoring, Analyse und Strategie 2022). Going for a walk individually was a legal outdoor activity even under the strictest lock-down regulations, which is why this type of action was chosen to circumvent the restrictions imposed on demonstrations (cf. ibid.).

On some occasions, these walks turned into gatherings in front of the private homes of politicians in charge for the COVID-19 measures on federal state level: In December 2021, around thirty members of the far-right group 'Freie Sachsen' (Free Saxonians) gathered in front of the home of Saxony's Minister of Health, Petra Köpping, carrying torchlights (cf. Klaus 2021). The incident received a lot of public attention, and was criticized in the media as well as by German politicians as attempts to intimidate politicians under the pretext of the protesters right to 'freedom of speech'. Also in December 2021, an investigative report by German journalists revealed that members of the Telegram chat group 'Dresden Offlinevernetzung' had talked about concrete preparations to assassinate Kretschmer, and only a few months later police foiled a plot to kidnap Germany's Minister of Health Karl Lauterbach and to sabotage power facilities in order to create chaos which would ultimately allow to overthrow the government. The incident made Germany's Interior Minister Nancy Faeser consider to close down the Telegram messenger service in Germany, which had been used for communication by the suspects (cf. Rogers 2022).

The messenger service Telegram in fact played a key role in the mobilization of the movement (cf. Forschungsinstitut Gesellschaftlicher Zusammenhalt 2020). The service allows the creation of large groups and channels to disseminate information quickly to a large number of users. Unlike Facebook or Google, which hand over data to authorities at their request, Telegram does not cooperate with authorities. This ensures a high degree of anonymity for its users which has proven beneficial for protest movements in a number of authoritarian countries such as Iran or Belarus. This anonymity poses a barrier to political repression and criminal prosecution. This makes the service attractive not only for social movements, but also for criminal and terrorist activities, and also facilitates the spread of hate-speech and calls for violence like those mentioned above.

The evolvement of the COVID-19 protests in Germany clearly demonstrate how online and offline contexts influence and often reinforce each other. Non-governmental organizations warned about the antidemocratic and violent potential of parts of the movement, and had pointed out the involvement of political activists from Germany's extremist far-right (cf. Bundesverband mobile Beratung 2020, Forschungsinstitut Gesellschaftlicher Zusammenhalt 2020). However, German authorities apparently underestimated the dangerous potential of this movement for a long time. Only after journalists had revealed the plans to murder Michael Kretschmer, and after several gatherings in front of private homes of politicians had taken place, the issue became more of a serious concern.

This points out the relevance of investigative work by critical journalists and non-governmental organizations to monitor the mobilization and networking strategies of far-right movements in Germany. Such monitoring often involves large amounts of data from the respective Telegram channels, often being manually analyzed which is enormously tedious. While machine learning-based approaches for such analyses exist, namely from the field of Natural Language Processing (NLP) , they are often not easily accessible for journalists and non-governmental organizations, making it hard for them to process large amounts of data from the often very dynamic contexts of informal political networks.

The open source project *Teledash* (Weichbrodt and Stanjek 2022) provides a research and analysis software for Telegram to overcome this gap. It allows to automatically download contents from one or more channels periodically, collects statistical data about the growth and activities of Telegram channels and groups, offers tools for text and speech recognition and a search function. Adding a functionality to extract meaningful parts of information from Telegram channels would be a beneficial contribution to the project.

The aim of my thesis is to explore the potentials and limitations of NER approaches for the extraction of named entities from German-language Telegram channels, focusing on the standard entity types person, location, and organization, and the non-standard entity types time, date, and action. I consider these entity types as particularly relevant for monitoring formal and informal protest activities.

My thesis is structured as follows: I will begin with a basic introduction to the task of NER, and outline the challenges related to it, both in general and with regard to user-generated texts such as Telegram messages. I will then present important shared tasks and datasets for German-language NER and NER in social media. As a next step, I will turn to state of the art approach for NER, namely transformers models as large pretrained language models which can be fine-tuned for specific NLP tasks such as NER. Having laid out the conceptual and theoretical foundations for my concern, I will then turn to existing NER solutions and qualitatively explore their applicability for my task. In the main part of my thesis, I will present my implementation of a custom model for NER in Telegram channels, and provide results from the experiments I conducted for this. As a final step, I will present first results from applying the custom model to example data from a German-language Telegram channel. For the sake of brevity and readability, I do not include extensive code parts in my thesis. Links to the corresponding notebooks are provided via footnotes in each section.

Chapter 2

Named Entity Recognition: An Introduction

NER is the task of extracting named entities such as for example names of persons, locations, or organizations from text. The task is situated within the field of NLP, and often an an important subtask of Information Extraction (IE) .

The origins of IE date back to the late 1970s, and its development was further accelerated in the late 1980s through funding by the US government for activities to evaluate IE technology. Organized by the Defense Advanced Research Projects Agency (DARPA) the first Message Understanding Conference (MUC) took place in 1987, providing standard datasets for international researchers to compete on. The MUC and was held seven times between 1987 and 1997. The first NER task at the MUC was introduced in 1996 and aimed at developing domain-independent systems for recognizing entities of types person, organization, and location, but also numerical expressions such as time, currency, and percentage (cf. Grishman and Sundheim 1996) in newspaper articles. In the following year, the organizers of the conference developed annotation guidelines which were used as a basis for several later NER tasks (cf. Chinchor 1997). Later activities such as the Automatic Content Extraction (ACE) meeting and the Text Analysis

Conference Knowledge Base Population (TAC KBP) shifted the focus to the extraction of more fine-grained entities, entity relations, and events, and later on to entity attribute extraction and entity linking. The provided datasets consisted of broader news data, covering political and international events, and later of open domain data, mainly from webpages (cf. Zong, Xia, and Zhang 2021, p. 229).

2.1 NER as a Sequence Labeling Task

NER is usually regarded as a sequence labeling task: To each word x_i in an input sequence, a label y_i is assigned, resulting in an output sequence Y which has the same length as the input sequence X (cf. Jurafsky and Martin 2021, p. 148).

Since named entities often consist of more than one token, the task actually requires labeling spans of texts, which makes it necessary to determine the correct boundaries of an entity in a text, and assign the correct label to each token in a detected span (cf. ibid., pp. 154–155).

NER thus involves the following two subtasks (cf.Zong, Xia, and Zhang 2021, p. 229):

1. **Entity detection**, to detect whether a given string in a text document is an entity, including the detection of the entity's boundaries.

2. **Entity classification**, to assign a specific category to the detected entity.

2.2 Standard Named Entity Types

The types of entities extracted via an NER system differ depending on the domain and aim of the application or task. For biomedical NER systems, the entity types of interest may for example be names of biological proteins or diseases. There are however standard entity types which are commonly the focus of general NER approaches. These are `PERSON`, `LOCATION`, `ORGANIZATION`, and sometimes `GPE` (geo-political entity). Often, an additional type is added which subsumes `OTH`er or `MISC`cellaneous named entities.

As we have seen above, earlier NER approaches also included numerical expressions such as time, date, currency/quantity, and percentage. Considering that a named entity is commonly defined as a single, unique real-world object, these numerical entities are not named entities in a narrow sense. They can, however, be useful categories depending on the purpose of an application (cf. Aggarwal 2018, p. 386).

2.3 The BIO Tagging Scheme

While some of the early NER approaches were exclusively rule-based, contemporary approaches usually rely on machine learning models trained on labeled datasets. For creating such datasets, an appropriate label set needs to be defined. A widely used tagging scheme is the BIO label set (cf. Ramshaw and Marcus 1995). Here, the B-tag is used to label the token representing the beginning of an entity (e.g. `B-PERSON)`, the I-tag is assigned to all of the following (intermediate or final) tokens of the respective entity (e.g. `I-PERSON`). All tokens who are located outside of named entity spans are labeled with the `O` tag (cf. Zong, Xia, and Zhang 2021, p. 231). The corresponding tag set thus consists of two labels per entity and the `O` tag, resulting in a total of $2n + 1$ tags, with $n = number\ of\ entity\ types$.

2.4 Evaluation of NER Systems

Similar to other NLP tasks, NER systems are usually systematically evaluated. This evaluation can be done in different stages of the development process, during deployment, and in production.

Evaluation in production usually involves application-specific business metrics, but also the involvement of stakeholders, and sometimes even end-users, which makes it rather expensive. Evaluation during development and deployment is therefore usually based on standardized machine learning metrics.

This intrinsic evaluation is less cost-effective since it can be done by the machine learning engineers themselves. Furthermore, it allows for adjustments and improvements in the earlier stages of the development process (cf. Vajjala et al. 2020, p. 71).

For supervised machine learning systems, the evaluation usually measures the model's performance on unseen data, called the test set (cf. Jurafsky and Martin 2021, p. 35). This test set is excluded from the training process and then used to evaluate the model. This is done by comparing the predictions of the model against the labels in the test set. (cf. Zong, Xia, and Zhang 2021, p. 241). The specific metrics used for intrinsic evaluation differ depending on the specific task. For NER, the standard metrics are F1, precision, and recall.

2.4.1 F1, Precision, and Recall

Precision and recall evaluate the model's predictions by taking into account the number of entities correctly recognized by the model as named entities (true positives, TP), the number of entities falsely tagged as entities by the model (false positives, FP), and the number of named entities not detected by the model (false negatives, FN). Precision and recall are defined as follows ((cf. ibid., p. 241):

$$precision = \frac{TP}{TP + FP} * 100$$

$$recall = \frac{TP}{TP + FN} * 100$$

Precision thus measures the percentage of correctly identified entities out of all entities detected by the system, i.e. how many correct predictions the model makes in its positive predictions. Recall measures the percentage of items tagged as named entities in the test set which are correctly recognized by the system, i.e. how many actual entities the model misses in its predictions.

The F1 score combines these two scores by computing a harmonic mean ((cf. Zong, Xia, and Zhang 2021, p. 241):

$$F1 = \frac{2 * precision * recall}{precision + recall}$$

F1 is the score commonly used in shared tasks on NER to evaluate the performance of the participating systems (cf. ibid., p. 241).

Compared to other sequence labeling tasks, NER poses certain challenges for evaluation: The fact that named entities can consist of multiple tokens brings up a segmentation problem. For evaluation this means for example that a system which recognizes only one token of a person's name consisting of two tokens will cause two errors: A false positive for tagging the second token as `O` instead of `I-PERSON`, and a false negative for not labeling the second token as `I-PERSON`) (cf. Jurafsky and Martin 2021, p. 167).

Past shared tasks for NER have applied different strategies to evaluate the participating systems. Earlier tasks relied on relaxed metrics, considering a prediction as correct if the correct label was assigned, regardless of the correct boundaries, or if the correct text boundaries were detected, regardless of the correct label (cf. Yadav and Bethard 2019). Later, strict metrics were introduced in which a prediction was only considered correct if both the predicted label and boundaries of an entity matched the true label and boundaries (cf. ibid.).

2.4.2 Accuracy

Accuracy is another common machine learning metric. It measures the percentage of items correctly classified by the system:

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} * 100$$

However, accuracy is not suitable for classification tasks on highly imbalanced datasets: In an NER dataset, the majority of tokens will be tagged as `O`. A classifier which would predict this majority class for all tokens without having learned anything from the training data would still achieve a very high accuracy (e.g. 90% accuracy for a dataset containing 90% `O` tags) (cf. Jurafsky and Martin 2021, pp. 65–66). This is why accuracy is not a suitable metric for NER.

2.5 General Challenges for NER

As an NLP task, NER faces challenges resulting from the characteristics of human language.

Ambiguity, i.e. uncertainty or inexactness of meaning, is inherent to most human languages. Resolving the ambiguity of words or phrases usually requires knowledge of the context in which they appear. For example, the word 'Bill' might refer to a persons first name, but also by a synonym for ìnvoice' when located at the beginning of the sentence. Making computer-based systems take into account and 'understand' **context** is one of the major challenges for NER (cf. Aggarwal 2018, p. 386).

Humans furthermore constantly use **common knowledge** in their communication, meaning that facts are assumed to be known and thus not expressed explicitly in statements. While every language is constituted and shaped by rules, it is far from being exclusively rule-driven. Rather, languages are **creative** and heterogenous, characterized by various styles, genres, and dialects.(cf. Vajjala et al. 2020, p. 14). Moreover, language evolves over time, in interaction with social and cultural processes, and the development of communication technologies such as social media. Existing datasets thus might not reflect new language phenomena such as hashtags. For named entities, this means that the set of known entities is never constant nor complete (Aggarwal 2018, p. 386).

Another challenge are **abbreviations** referring to multiple instances of the same named entity, such as 'U.S.', 'USA' or 'United States of America', all referring to the same geopolitical entity. Entity disambiguation is thus usually an important follow-up task for NER(cf.Aggarwal 2018, p. 387).

Capitalization conventions vary widely across languages: In English-language texts for example, capitalization usually indicates proper names and can thus be used as a feature for NER. In German by contrast, all nouns are capitalized, which makes the feature less helpful (cf. Benikova, Biemann, and Reznicek 2014).

Diversity across languages makes it difficult to transfer a language-specific application to other languages. Building language-agnostic solutions is a very complex matter, while building separate solutions for each language is labor- and time-intensive (cf. Vajjala et al. 2020, p. 14).

Imbalanced resources for different languages limit the diversity both of research and applications. In fact, most NLP technologies are still designed for English or other European languages (cf. Singh 2008). Correspondingly, a large number of datasets and libraries exists for these high-resource languages. Low-resource languages are usually significantly 'less studied, resource scarce, less computerized' (Magueresse, Carles, and Heetderks 2020). This imbalance is also reflected in existing datasets for NER.

2.6 Challenges for NER in User-Generated Texts

While the above-mentioned challenges apply in general for all NER approaches, the characteristics user-generated texts from online domains further complicate the task for classic machine learning based NER approaches.

Social media texts are often **more informal and noisy** than newspaper articles, they contain misspellings or domain-specific expressions such as hashtags. Figure 2.1 illustrates the noisy character of tweets as an example.

Moreover, user-generated texts from social media are often of **shorter length** than e.g. newspaper articles which might result in a lack of context (cf. Ritter et al. 2011).

The **wide variety of capitalization styles** are another challenge, like e.g. the use of only lowercase for German texts, or the use of uppercase only in some texts (cf. Ritter et al. 2011). As ibid. point out, earlier news-trained NER systems seemed to rely heavily on capitalization which limited their applicability to user-generated social media texts.

Also, these texts usually contain more out of vocabulary (OOV) words than e.g. news articles. This is partly due to misspellings, but also stems from platform specific codes and styles: The frequent occurence of **special characters** such as # for hashtags or for user mentions poses challenges because they cannot easily be handled during annotation and training. User-generated text in social media also often includes a lot of **lexical variations**. ibid. collected more than fifty variations for the word ‘tomorrow’ from a Twitter corpus, as shown in figure 2.2.

1	The Hobbit has FINALLY started filming! I cannot wait!
2	Yess! Yess! Its official Nintendo announced today that they Will release the Nintendo 3DS in north America march 27 for $250
3	Government confirms blast n nuclear plants n japan...don’t knw wht s gona happen nw...

Figure 2.1: Examples of noisy text in tweets, taken from Ritter et al. 2011

‘2m’, ‘2ma’, ‘2mar’, ‘2mara’, ‘2maro’, ‘2marrow’, ‘2mor’, ‘2mora’, ‘2moro’, ‘2morow’, ‘2morr’, ‘2morro’, ‘2morrow’, ‘2moz’, ‘2mr’, ‘2mro’, ‘2mrrw’, ‘2mrw’, ‘2mw’, ‘tmmrw’, ‘tmo’, ‘tmoro’, ‘tmorrow’, ‘tmoz’, ‘tmr’, ‘tmro’, ‘tmrow’, ‘tmrrow’, ‘tmrrw’, ‘tmrw’, ‘tmrww’, ‘tmw’, ‘tomaro’, ‘tomarow’, ‘tomarro’, ‘tomarrow’, ‘tomm’, ‘tommarow’, ‘tommarrow’, ‘tommoro’, ‘tommorow’, ‘tommorrow’, ‘tommorw’, ‘tommrow’, ‘tomo’, ‘tomolo’, ‘tomoro’, ‘tomorow’, ‘tomorro’, ‘tomorrw’, ‘tomoz’, ‘tomrw’, ‘tomz’

Figure 2.2: Lexical variations of the word ‘tomorrow’ in tweets, taken from Ritter et al. 2011

Last but not least, social media texts often **mixed-language texts**, making it harder for language-specific models to accurately determine named entities in them.

Given these differences between social media texts and more formal genres, NER on social media needs to handle the problem of **domain adaptation**. Most of the existing NER systems are in fact trained on a small selection of standard corpora mainly consisting of newswire texts, and earlier studies found that applying such a system to social media texts resulted in a drop of performance (cf. Ritter et al. 2011). This might also be true for the application of existing solutions to Telegram messages.

Chapter 3

Related Work

To the best of my knowledge, there are no works using NER for monitoring the COVID-19 protests in Germany, or any other works focussing on NER in German-language texts from messenger services. There are a few works using NER for analyzing (English-language) discourses in the context of the COVID-19 pandemic: Tangherlini et al. as well as Shahsavari et al. used NER as a component for analyzing conspiracy narratives and the corresponding communities in blog texts and news articles (Tangherlini et al. 2020), and in comments from the imageboard 4chan (Shahsavari et al. 2020).

There is, however, a number of shared tasks and corresponding datasets contributed to the development of German-language NER and NER in social media text data. For NER in social media texts, I will also present a selection of experimental studies which formed the basis for later shared tasks.

3.1 Shared Tasks on German-Language NER

A first important cornerstone for German-language NER was the Conference on Computational Natural Language Learning (CoNLL) for English and German-language NER in 2003. The task focused on the entity types person, location, organization, and miscellaneous. The provided German dataset consists of 909 newspaper articles from the Frankfurter Rundschau.

The dataset was annotated by non-native speakers at the University of Antwerp which is assumed to have led to annotation inconsistencies.[1] The best participating system was a system combining Maximum Entropy and Hidden Markov Models, and achieved an F1 score of 72.41 for the German-language dataset (cf. Tjong Kim Sang and De Meulder 2003)

In 2014, GermEval, a series of shared tasks on Natural Language Processing for German, announced an NER task. The provided dataset consists of 31,000 sentences sampled from Wikipedia articles and online news, and includes fine-grained annotations for entities of type person, organization, location, and other. The Germeval 2014 corpus includes nested and derived entities and is thus not readily combinable with other datasets. While the corpus consists of online resources, it does not contain noisy user-generated texts. The best system achieved an F1 score of 79.08 for first-level named entities, using character query-based features in order to find sequences that are characteristic for named entities, and vocabulary clusters for semantic generalization (cf. Benikova, Biemann, and Reznicek 2014).

The two corpora represent standard datasets for German-language NER and are used until today to train and evaluate state of the art NER models.

3.2 Experimental Studies and Shared Tasks on Social Media NER

While the first shared tasks on NER focussed on newswire articles, later work began to explore NER in user-generated texts from social media. Research on NER for social media texts is almost completely focused on English-language texts: In 2017, Bhoi and Balabantaray analyzed seven different systems for NER in social media text, out of which five were exclusively English. The only system supporting German language was the Stanford NER tagger, which was trained on newswire, not on social media texts though. The

[1]Some authors, e.g. Schweter and Akbik 2020 mention a revised edition, but I was not able to find further details or the mentioned resource itself.

authors compare system performance for tweets and microblog content. They report best F1=0.686 for tweets, and best F1=0.889 for the more formal microblog content, concluding that 'state-of -the-art NER methods are giving unsatisfactory results in that informal and noisy content of social media text' (Bhoi and Balabantaray 2017)

Ritter et al. were among the first to explore NER in social media texts. In 2011, the authors conducted an experimental study, using a random sample of 2,400 English-language tweets annotated with ten different entity types[2]. The authors use a discriminate approach for segmentation, and a weakly supervised approach for classification, relying on topic modeling and a large knowledge base. The system achieved an F1 score of 0.66 when applied to ten entity types. When constrained to the three standard entity types person, location, and organization, the F1 score increased to 0.75, with 0.83 for person, 0.74 for location, and 0.66 for organization (cf. Ritter et al. 2011).

Another study conducted in 2015 by Derczynski et al. 2015, using three different English-language datasets including the Ritter et al. corpus, confirmed the rather low performance of by then existing machine-learning based NER on noisy user-generated data: The evaluated systems achieved F1 scores ranging between 40.27, 51.49, and 77.18 on the different datasets (cf. ibid.). In line with Ritter et al. 2011, the authors identify unreliable capitalization, typographic errors, abbreviations, and lack of context due to the shortness of texts as main challenges.

Also in 2015, Baldwin et al. organized a shared task for named entity recognition on social media data, namely tweets from Twitter in the context of the Workshop on Noisy User-generated Text (W-NUT), using the Ritter et al. corpus. The best participating system combined the Stanford NER tagger with entity linking via a knowledge base. It achieved an F1 score of 56.41 (cf. Baldwin et al. 2015). The W-NUT series continued to address the task of NER in subsequent years: Tasks were organized on named entity recognition in tweets (cf.Strauss et al. 2016), and novel and emerging entity recognition

[2]Including person, geo-location, company, product, facility, tv-show, movie, sportsteam, band, and other (cf. Ritter et al. 2011)

(cf. Derczynski et al. 2017), both using the Ritter corpus.[3]. The achieved F1 scores remained rather low with best F1=52.41 in 2016, and best F1=41.86 in 2017.

Even though NER on social media texts remained a challenging task regarding these results, W-NUT shifted its focus to other tasks and domains in subsequent years. In parallel, larger human-annotated social media NER datasets have been created (cf. Derczynski, Bontcheva, and Roberts 2016). However, there is still a lack of such corpora for languages other than English.

While the introduction of transformer models in recent years has outperformed prior approaches in many NLP tasks including NER, they are commonly fine-tuned and evaluated on the standard CoNLL and Germeval 2014 corpora.

[3]In 2017 the Ritter corpus was used as training set, extended by ~1.000 manually annotated documents from different online forums and platforms as development set, and ~ 1.300 documents manually annotated Youtube comments as test set

Chapter 4

State of the Art NER with Transformer Models

Since their beginnings in the 1970s, NER approaches have evolved mostly in parallel to general advancements in NLP: Starting with early rule-based and knowledge-based systems, the field moved to machine-learning based approaches, often involving extensive feature-engineering. Deep-learning based approaches using neural networks and word embeddings then began to replace these architectures, making feature-engineering more and more obsolete, and allowing for richer forms of word representations. (cf. Yadav and Bethard 2019). Most recently, transformer models based on self-attention have achieved new state of the art performance for several NLP tasks including NER.

The transformers architecture was first introduced in 2017 by Vaswani et al., in the seminal paper 'Attention Is All You Need' (Vaswani et al. 2017).

Like prior appraoches, transformers can be used to map input sequences to output sequences of the same length (cf. Tunstall et al. 2022, p. 3). However, they represent a clear shift from prior model architectures since they do not process their inputs sequentially. This allows for a parallelization of computations, and furthermore represents a new way to consider the context of input words.

4.1 Self-Attention

Transformer models use a mechanism called **self-attention** which allows them to represent each word with respect to its current context (cf. Vajjala et al. 2020, pp. 25–26). Through self-attention, these models are able to use information from (theoretically) arbitrarily large contexts, without having to pass this information sequentially, for example through recurrent connections such as in Recurrent Neural Networks (RNNs) (cf. Jurafsky and Martin 2021, p. 191).

Attention in transformer models can be conceptualized as '[comparing] an item to a collection of other items in a way that reveals their relevance in the current context.' (ibid., p. 192). Self-attention then means to compare a given item to other elements within a sequence (cf. ibid., p. 192).

When processing an input item, a backward looking self-attention layer has access to the item itself and all of the inputs before it, but not to the succeeding inputs. In bidirectional architectures such as BERT, the considered context is not limited to the preceding inputs (see Devlin et al. 2018 for further explication). The computation for each item in a self-attention layer is carried out independently from other input items, which allows to run a high number of computations in parallel (cf. Jurafsky and Martin 2021, p. 192. Transformer models thus allow for parallelization which is an advantage in terms of computational speed and efficiency compared to prior sequential approaches such as Conditional Random Fields (CRFs), RNNs, or Long Short Term Memory Networks (LSTMs).

4.2 Key, Query, and Values

The attention mechanism of transformer models consider each input in three different roles (cf. ibid., p. 193).

- As a **query**, the input represents the current focus of attention and is compared to all of its preceding inputs.
- If the input is among the preceding inputs (given a unidirectional, left-to-right model architecture), it is compared to the current query, i.e. the current focus of attention. In this case, it represents a **key**.
- Whenever we use the input for computing the current output, it is referred to as **value**.

Figure 4.1 illustrates the process of calculating the value for the third element in a sequence including the key, query, and value vectors for the current item x_3 and the preceding items x_1 and x_2. We can see that after generating these vectors, they key-query comparisons are carried out, which are then passed to the softmax function. The resulting values are then multiplied by the value vector, and summed up to obtain the output vector y_3.

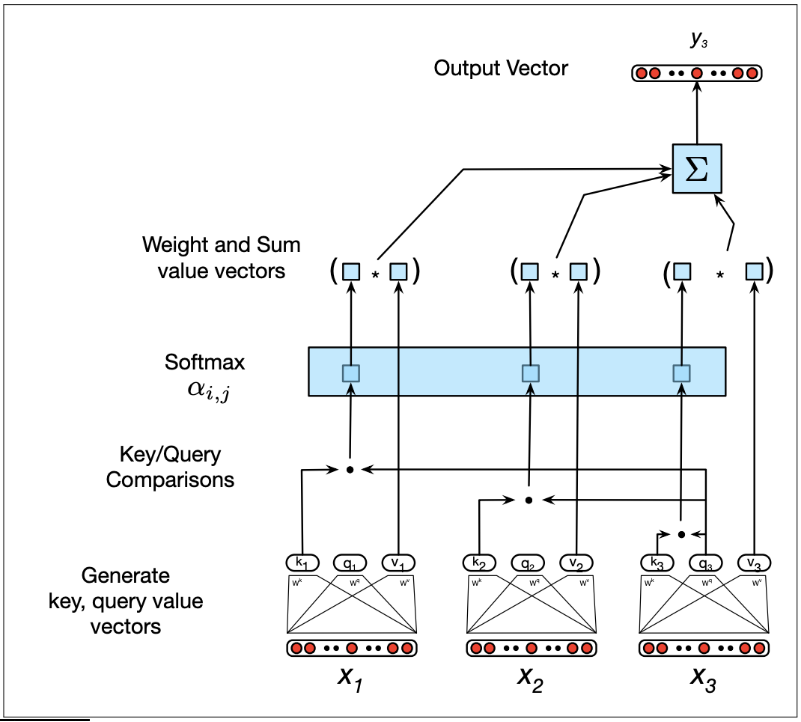


Figure 4.1: Calculating the value y_3 for the third element x_3 in a sequence, using left-to-right-attention; taken from Jurafsky and Martin 2021, p. 194

4.3 Multi-Head Attention

They way context is conceptualized and modelled in this architecture is thus fundamentally different compared to prior approaches. Furthermore, by extending the concept of self-attention to multiple heads, transformer models are able to consider different aspects of relationships between input words. This mechanism is called multi-head attention, and implemented in the form of multiple, parallel self-attention layers which are located at the same depth of the network. Each of these heads learns its own parameters, allowing it to pay attention to different ways in which the words in the input relate to each other (cf. Jurafsky and Martin 2021, p. 194).

4.4 Transformer Blocks

Transformer models are composed of so-called transformer blocks. All of them have the same input und output dimensions, which allows to stack them on top of each other (cf. ibid., p. 194).

These transformer blocks are themselves multilayer networks, containing the self-attention layers, simple linear layers, and feedforward networks. As a mechanism to improve training performance, residual connections and layer normalization are implemented in each block, allowing to pass information directly to the next layer, and keeping the output values of each hidden layer in a certain range (cf. ibid., p. 194).

There are, however, some specific limitations of transformer models:

4.5 Limited Input Length

The way attention is computed makes the process in transformer models expensive for long inputs: For every input, each pair of tokens needs to be compared. Against this background, most transformer models are limited to a certain input length, e.g. 512 tokens for BERT models, to keep computational cost within acceptable bounds.(cf. ibid., p. 195).

4.6 Positional Embeddings

Furthermore, since transformer models do not work sequentially, the information about the positional context of each item in the input is lost. To provide this information, transformer models use additional positional embeddings to include this information. These embeddings are learned during training, just like the word embeddings, and encode information about the context for each input item (cf. Jurafsky and Martin 2021, p. 195).

Figure 4.2 shows the original transformer architecture introduced by Vaswani et al. 2017, with the insertion of positional embeddings (positional encodings) before the multi-head attention layers.

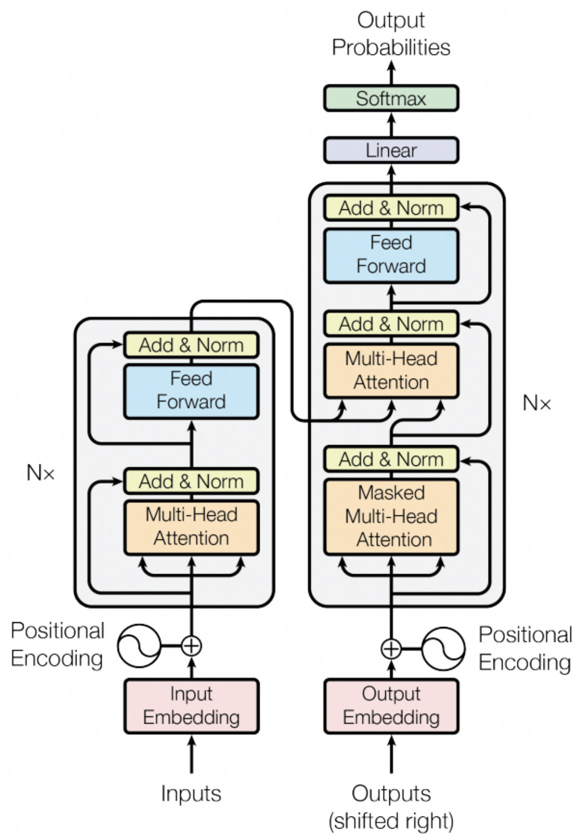


Figure 4.2: Transformer model architecture as introduced by Vaswani et al. 2017

4.7 Model Architectures

There is a large number of different architectures for transformer models, resulting in an even larger number of available models.[1]

The three main architectures for transformer models are encoders, decoders, and encoder-decoders. Encoder models are still the most commonly used, both in research and for business applications (cf. Tunstall et al. 2022, p. 79). These general architecture types can be further differentiated into different model architectures such as e.g. DistilBERT or XLM models (see ibid., p. 79 for an overview and ibid., pp. 57–86 for details on the different architectures).

4.8 Pre-Training and Fine-Tuning of Transformer Models

Transformers are usually trained as large language models on different tasks such as Next Sentence Prediction (NSP) or Masked Language Modeling (MLM). Such pretrained language models can then be fine-tuned for downstream tasks such such as NER. This is achieved by first training a language model on large amounts of unlabelled text data, which is called pre-training. Pre-training allows the model to incorporate rich information about language in general, which allows it to easier learn specific tasks, called downstream tasks, on smaller datasets later. This second step is called fine-tuning. Fine-tuning requires labeled data, but can be achieved with datasets of much smaller size compared to classic supervised machine learning approaches, since the model has already achieved a lot of general knowledge about a given domain during pre-training (ibid., pp. 6–8).

I will explore the potentials of fine-tuning transformer models for my purpose in chapter 6 in practice. But before, I will take a closer look at existing solutions for NER.

[1]By the time of writing this thesis, the Hugging Face Hub lists more than 62.000 of pretrained models available for download: `https://huggingface.co/models`, accessed on 2022/08/06.

Chapter 5

SpaCy and Flair: A Qualitative Exploration of Existing Solutions

Most of the existing NER models are limited to the four common entity types PERSON, LOCATION, ORGANIZATION, and OTHER/MISC. To the best of my knowledge, there are no existing models supporting the detection of the entity types TIME, DATE, and ACTION. An exception is the English-language spaCy model which supports TIME and DATE recognition.

In order to explore the applicability of existing models for my purpose, I selected some sample messages from a German-language Telegram channel and qualitatively evaluated the results of two German-language NER models. Since the English spaCy model supports more entity types, and English models are known to be well-resourced, I used machine translation to translate the sample texts into English and explore the results of the English models as well.[1]

All tested models were selected from the library spaCy and the framework

[1] For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/3_exploration/spacy_flair_bert_first_experiments`.

FLAIR. The models were chosen due to their easy applicability, allowing for fast exploration and analysis, and because both of them support German- and English-language named entity recognition. I am aware of the fact that the observations presented here cannot be generalized without quantitative evaluation on a larger sample.

5.1 Example Data

I randomly selected ten example messages from the mainly German-language Telegram channel 'Demotermine' (*Demo dates*). Prior qualitative exploration of sample messages indicated that this channel contains a lot of messages mobilizing for various kinds of protests against the COVID-19 containment measures in Germany. This makes it a suitable source for my purpose.

5.2 German-language NER with spaCy and Flair

My first analysis focusses on results for German-language sample texts, using the medium-size German-language spaCy model and the standard German-language Flair NER model.

5.2.1 The German-language spaCy model

spaCy is a free open-source library for NLP. The library is extensively documented and thus serves as an easy starting point for beginners, while also claiming to be designed for use in production. spaCy supports more than 66 languages (including German) and supports a variety of NLP tasks such as part-of-speech tagging, text classification, and NER (cf. ExplosionAI 2022b). The medium-size German model supports to the common entity types `LOC`, `MISC`, `ORG`, and `PER` (cf. ExplosionAI 2022c).

The central component of the model's NER pipeline is spaCy's Entity Recognizer based on transition-based algorithm which assumes that the most relevant information about an entity is close to its first token (cf. ExplosionAI 2022a). The underlying TransitionBasedParser is a neural network state prediction model which embodies 'an approach to structured prediction where the task of predicting the structure is mapped to a series of state transitions.' (ExplosionAI 2022d).

The model was trained on the WIKINER corpus, a free, multilingual corpus created from Wikipedia articles. The annotations were created automatically (cf. Nothman et al. 2013). The corpus is thus only silver standard (compared to gold standard corpora based on human annotations). The model achieved an F1 score of 0.84 (cf. ExplosionAI 2022c).

5.2.2 The German-language Flair Model

The FLAIR framework is an NLP library developed to 'facilitate training and distribution of state-of-the-art sequence labeling, text classification and language models' (Akbik et al. 2019). The overall aim is to provide an interface for the easy use of different word and document embeddings (cf. ibid.).

The German-language Flair model for NER uses a combination of GloVe embeddings (cf. Pennington, Socher, and Manning 2014 and Flair embeddings (cf. Akbik, Blythe, and Vollgraf 2018) to create word embeddings. Flair embeddings create 'contextual string embeddings' (ibid.) which generate different embeddings for the same word depending on its context in a document (cf. ibid.). The word embeddings passed to a Bidirectional Long Short Term Memory (BiLSTM)-CRF sequence labeling model for the NER downstream task (cf. ibid.).

I used Flair's German standard 4-class NER model, supporting the entity types person, location, organization, and other. It was trained on the CoNLL 2003 corpus, reaching an F1 score of 87.94 (cf. Akbik, Blythe, and Vollgraf 2022b).

5.2.3 Results

Table 5.1 shows the results for both models out of which the following observations can be made: Overall, the spaCy model yields more false positives: It falsely tags the sequences 'Endkundgebung' (final rally), 'GEHT DEMONSTRIEREN' (go demonstrate), 'Dresdner &', 'Videos', 'Berliner', 'Versammlungsfre' (freedom of assembly, truncated), 'Einfach' (simple/simply), 'Der Mann mit dem Schild' (the man with the sign), and 'Einsatzwägen' (patrol cars).

The Flair model only falsely tags 'FU' and 'Demo-Berlin'. The sequence 'FU' does not exist in the original text, but is apparently a result of the model's pre-processing. Tagging 'FU' as organization is plausible however, as is tagging 'Demo-Berlin' as location since the sequence contains 'Berlin'.

The spaCy model sometimes correctly detects a named entity, but then assigns the wrong tag to it: 'Blankenfelde Mahlow' represents a location, but is tagged as person.

The spaCy model does not detect 'Freiburg' in the first message and thus yields a false negative, while the Flair model correctly detects the location mention.

Somewhat surprisingly, spaCy recognizes parts of the sequence 'Platz der alten Synagoge' (square of the old synagogue) as a location which is missed by the Flair model.

Even though the sample is too small for generalization, my observations casts strong doubts on the claim that spaCy offers a production-ready NER solution. I assume that the differences in performance result both from the different model architectures and the quality of the respective training corpora.

Table 5.1: Detected entities for German-language example messages

No.	Text	spaCy/de_core_news_md	flair/ner-german
1	Freiburg Aufzug mit Endkundgebung am Platz der alten Synagoge . Live Musik ! Details 19062021 Raus auf die Straße	Endkundgebung[LOC], alten Synagoge[LOC]	Freiburg[LOC]
2	GEHT DEMONSTRIEREN ! - Message von Elijah Tee an die Dresdner & anderen Demonstranten in Deutschland Morgen am 13.06. um 16.30 Uhr und bundesweit überall , wo es Mitdenker gibt . - Bilder und Videos hochladen - Diskussion Fahren oder Mitfahren	GEHT DEMONSTRIEREN [ORG], Elijah Tee[PER], Dresdner &[LOC], Deutschland[LOC], Videos[MISC]	Elijah Tee[PER], Deutschland[LOC]
3	15831 Blankenfelde Mahlow Berliner Speckgürtel Ausfahrt Lichtenrade am 14.03.2021 14032021 Raus auf die Straße	Blankenfelde Mahlow[PER], Berliner[MISC]	Blankenfelde Mahlow[LOC], Lichtenrade[LOC]
4	2021 10 11 Sven Liebich fragt Ministerpräsident Haseloff nach dessen Verständnis von Versammlungsfre - Sven Liebich 0:23 Raus auf die Straßen Übersicht / Overview	Sven Liebich[PER], Haseloff[PER], Versammlungsfre[PER], Sven Liebich[PER]]	Sven Liebich[PER, Haseloff[PER], Sven Liebich[PER]
5	FÜR ALLE DEMOKRATEN GEFAHR : Die Polizei blockiert den Stern ! Deswegen laufen die Menschen vom Ernst-Reuther-Platz nach Südosten zum Breitscheitplatz und dort weiter auf den Kudamm ! ! Schließt euch an : Neue Kantstraße direkt Richtung Osten auf den Breitscheitplatz ! Von der Straße des 17. Juni Richtung Süden zum Kudamm !	DEMOKRATEN[ORG], Ernst-Reuther-Platz[LOC], Breitscheitplatz[LOC], Kudamm ! ![LOC]), Neue Kantstraße[LOC], Breitscheitplatz[LOC], Kudamm ![LOC]	FU[ORG], Ernst-Reuther-Platz[LOC], Breitscheitplatz[LOC], Kudamm[LOC], Kantstraße[LOC], Breitscheitplatz[LOC], Kudamm[LOC]
6	Dr. Gunter Frank : Wie Moralismus dazu führt das man Feinde sieht und vernichten will — Antihygienista - Offene Gesellschaft Kurpfalz 15:04 Raus auf die Straßen Übersicht / Overview	Gunter Frank[PER], Antihygienista[LOC], Offene Gesellschaft Kurpfalz[ORG]	Gunter Frank[PER], Antihygienista[ORG], Offene Gesellschaft Kurpfalz[ORG]
7	Einfach nur Krank solche Politiker ! Was sollen wir mit ihm machen wenn unsere Kinder anhand dieser Impfung nen schaden erleiden ? Wer ist dann dafür verantwortlich , wen schmeißen wir dann in den Knast und nehmen sein ganzes Geld weg , sollen wir das mit dir machen ? Raus auf die Straße	Einfach [MISC]	*no entities detected*
8	27.11.2021 Frankfurt Durchsagen von gemischt mit Musik . Super Idee Für mehr Berichte folgen auf ... Raus auf die Straßen Übersicht / Overview	Frankfurt[LOC]	Frankfurt[LOC]
9	Demo-Berlin : Der Mann mit dem Schild wurde soeben von der Polizei festgenommen . Circa 7 Einsatzwägen sind vor Ort und umzäunen den Park großräumig . Am Humboldthain gibt es nichts mehr zu gewinnen . # b2908 Raus auf die Straßen	Der Mann mit dem Schild[LOC], Einsatzwägen[LOC], Humboldthain[LOC]	Demo-Berlin[LOC], Humboldthain[LOC]
10	Anthony Fauci : Definition von " vollständig geimpft " könnte geändert werden — Epoch Times Nachrichten 6:59 Raus auf die Straßen Übersicht / Overview	Anthony Fauci[PER], Times Nachrichten[MISC]	Anthony Fauci[PER], Epoch Times Nachrichten[ORG]

5.3 Exploring the Potentials of Machine Translation for NER

Even though the German spaCy model performed worse than the German Flair model, its English equivalent comes with one great advantage for my purpose: It supports a greater number of entity types, including DATE and TIME, making it potentially useful for monitoring mobilization for protests. Furthermore, regarding the fact that English models are generally known to be more well-resourced, my aim was to explore the performance of the English variant for both models.

5.3.1 The English-language spaCy Model

The English-language spacy model supports a total of 18 entity types, including the standard entity types person, location, organization. For my aim, the entity types time and date are particularly relevant. For a detailed overview of all entity types and their definitons see table A.1 in the appendix.

The underlying architecture is the same as for the German model.

Judging by the listed data sources in the documentation, I assume that NER model was trained on the OntoNotes 5 corpus, containing data from various text genres including news, phone conversations, blogs, usenet newsgroups, etc. (cf. Ralph Weischedel 2013). The model achieved an F1 score of 0.85.

5.3.2 The English-language Flair Model

The English-language Flair model for NER supports the same four standard entity types as the German version. Unlike the English spaCy model, it offers no advantages in terms of entity types.

The architecture of the standard English-language Flair model is the same as for the standard German model described in section 5.2.2.

The English model was trained on a corrected version of the CoNLL 2003 corpus and achieved an F1 score of 93.06 (cf. Akbik, Blythe, and Vollgraf

2022a).

For implementation, the example messages were first translated using the DeepL API.[2]. The implementation for obtaining the NER results from both models followed the same procedure as for the German models.[3]

5.3.3 Results

Table 5.2 shows the results of both models for the translated texts.

We can observe that the overall performcance of the spaCy model deteriorates compared to the German version: Out of the 25 entity tags, 17 are misclassifications, e.g. the person name 'Sven Liebich' as organization, or the location 'Humboldthain' as person, to name just a few. Furthermore, it misses named entities which were recognized by its German equivalent, e.g. the person mentions 'Haseloff' and 'Elijah Tee'. The model also proves largely unsuitable for extracting dates and times: Misclassifying e.g. the date '13.06' or the time mention '16.30' as cardinals, and the postal code '15831' as date. Out of the 11 mentions of times or dates in the example data, it correctly recognizes only the sequence '03/14/2021 14032021' in message no. 3 as date.

The performance of the English Flair model worsens as well compared to its German equivalent, although not quite as strong as the spaCy model: The model yields more false entity classifications, e.g. of 'Raus auf die Straßen Overview' as organization, or 'Dresdeners', 'Streets Overview', 'Stern!' as miscellaneous. Regarding locations, the model now misclassifies the mentions in message no. 3 as miscellaneous, which were at least partly correctly tagged by the German Flair model. Interestingly, it is the only model which partly recognizes the location mention 'Straße des 17. Juni'.

[2] https://www.deepl.com/docs-api. For implementation details of the translation step, see: https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/3_exploration/translation

[3] For NER implementation, see https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/3_exploration/spacy_flair_bert_first_experiments.

Table 5.2: Detected entities for English-language example messages

No.	Text	spaCy/en_core_web_md	flair/ner-english
1	Freiburg procession with final rally at the square of the old synagogue . Live music ! Details 19062021 Out on the street	*no entities detected*	Freiburg[LOC]
2	GO DEMONSTRATE ! - Message from Elijah Tee to the Dresdeners & other demonstrators in Germany Tomorrow on 13.06. at 16.30 clock and nationwide everywhere , where there are fellow thinkers . - Upload pictures and videos - discussion drive or ride along	Germany[GPE], Tomorrow[DATE], 13.06[CARDINAL], 16.30[CARDINAL]	Elijah Tee[PER], Dresdeners[MISC], Germany[LOC]
3	15831 Blankenfelde Mahlow Berlin Speckgürtel Lichtenrade exit on 03/14/2021 14032021 Out on the road	15831[DATE], Blankenfelde Mahlow Berlin[PERSON], Speckgürtel Lichtenrade[PERSON], 03/14/2021 14032021[DATE]	"Blankenfelde Mahlow Berlin Speckgürtel Lichtenrade"[MISC]
4	2021 10 11 Sven Liebich asks Prime Minister Haseloff about his understanding of assembly free - Sven Liebich 0:23 Raus auf die Straßen Overview / Overview	Sven Liebich[ORG], Haseloff[PER], Straßen Overview / Overview[ORG]	Sven Liebich[PER], Haseloff[PER], Sven Liebich[PER], Raus auf die Straßen Overview[ORG]
5	FOR ALL DEMOCRATS DANGER : The police blocks the Stern ! That's why people run from Ernst-Reuther-Platz to the southeast to Breitscheitplatz and there further to Kudamm ! ! Join : Neue Kantstraße directly to the east to Breitscheitplatz ! From the Straße des 17. Juni southwards to the Kudamm !	DEMOCRATS[NORP], Ernst-Reuther-Platz[ORG], Breitscheitplatz[PERSON], Kudamm[FAC], Neue Kantstraße[PERSON], Straße[GPE], 17[CARDINAL], Kudamm[FAC]	Stern ![MISC], Ernst-Reuther-Platz[LOC], Breitscheitplatz[LOC], Kudamm[LOC], Neue Kantstraße[ORG], Breitscheitplatz[LOC,]Straße des[LOC], Juni[LOC], Kudamm[LOC]
6	Dr. Gunter Frank : How moralism leads to the fact that one sees enemies and wants to destroy - Antihygienista - Offene Gesellschaft Kurpfalz 15:04 Raus auf die Straßen Übersicht / Overview	Gunter Frank[PER], 15:04 Raus auf[PER], Straßen Übersicht / Overview[ORG]	Gunter Frank[PER]
7	"Just sick such politicians ! What should we do with him if our children suffer from this vaccination ? Who is responsible for it , who do we throw in jail and take away all his money , shall we do that with you ? Out on the street"	*no entities detected*	*no entities detected*
8	27.11.2021 Frankfurt announcements of mixed with music . Super idea For more reports follow on ... Out on the streets Overview	Frankfurt[GPE]	Frankfurt[LOC]
9	Demo-Berlin : The man with the sign has just been arrested by the police . About 7 emergency vehicles are on site and fence the park extensively . At the Humboldthain there is nothing more to win . # b2908 Out on the streets	Humboldthain[PER], #[CARDINAL]	Humboldthain[LOC]
10	Anthony Fauci : Definition of " fully vaccinated " could be changed - Epoch Times News 6:59 Out on the Streets Overview / Overview.	Anthony Fauci[PER], the Streets Overview / Overview[ORG]	Anthony Fauci[PER], Epoch Times News[ORG], Streets Overview[MISC]

5.4 Summarization

I selected ten example messages from the Telegram channel ‘Demotermine’, containing mobilization messages for protests against the COVID-19 measures in Germany. I applied two existing NER models, the German-language spaCy model and the German-language Flair model for NER and explored the named entities recognized by them. I then translated the example messages to English, and applied the English-language equivalents of the two models to them to explore the potentials of machine translation for my purpose.

My findings indicate that the existing solutions are only of limited use for my purpose: The predictions of the spaCy model contain a lot of misclassifications, both for German and English. The Flair model delivers more accurate results, with a decrease in performance for the English data. One reason for this could be that the person or location names are language area specific, and therefore better recognized by models trained on data from their origin language area. Most importantly, most of the analyzed models are restricted to the standard entity types person, location, organization, and other/miscellaneous. This makes it impossible to extract e.g. time or date mentions which are of central relevance for monitoring COVID-19 protests. The only model which provides these entity types delivers poor results and misses by far most of the date and time mentions included in the examples. Against this background, the remaining part of my thesis explores the potentials and limitations of building a custom solution.

Chapter 6

Building a Custom Model for NER in Telegram Channels

As we have seen in the previous chapter, existing solutions are not applicable for my purpose. Hence I decided to train a custom NER model NER model for extracting the entity types person, location, organization, time, date, and action from Telegram messages. For this, I fine-tuned a selection of pretrained transformer models using the Hugging Face Transformers library.[1]. Since fine-tuning for NER requires labeled data, I created a detailed annotation scheme and annotated a small-size Telegram corpus of 500 messages. As additional data, I used the Smartdata corpus, a dataset consisting of 2.598 documents created for event extraction in online texts. I evaluated five different models on the two datasets separately, and on the combined corpus. Finally, I selected the best model regarding a combined assessment of its F1, precision and recall scores.

6.1 Building the Telegram Corpus

The small-size Telegram corpus was created from the same channel I used for my exploration of existing models in chapter 5: The channel 'Demotermine',

[1] `https://huggingface.co/docs/transformers/`

disseminating mainly messages mobilizing for protests against COVID-19 measures in German language.

6.1.1 Raw Data

The raw data covers a time period from March 3rd, 2018 when the channel was created, until December 19th, 2021. By the beginning of June 2022, the channel had around 52.900 subscribers.

The channel thus existed long before the COVID-19 pandemic. However, an analysis of message frequencies over time clearly shows the channel gained significant momentum in spring 2020, at the beginning of the pandemic in Germany (see figure 6.1).

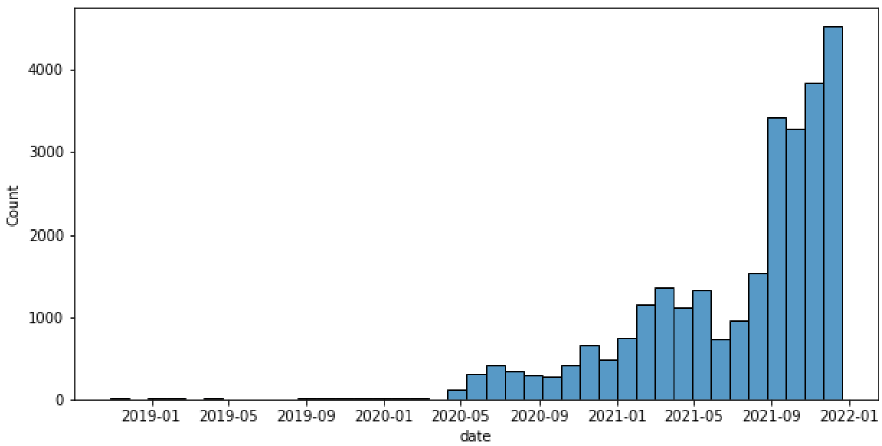


Figure 6.1: Message frequencies in Telegram channel 'Demotermine'

6.1.2 Text Cleaning

Text cleaning was restricted to an absolute minimum to make the corpus reflect a realistic sample of user-generated texts. Besides the removal of newline characters, compound hashtags were split up into separate hashtags (e.g. transforming #food#delicious into #food #delicious).

@user_mentions and *#hashtags* were included as whole sequences, same as sequences containing in-word hyphens. URLs and emojis were included as well. The capitalization of words was retained. For tokenization, the tokenizer of the German medium-size spaCy model was applied with a customized token match pattern.[2] Text cleaning resulted in empty rows for some documents. After removing these emtpy rows and text duplicates, 27,627 messages remained for sample selection.

6.1.3 Sample Selection

The following steps were carried out for sample selection:[3] To be able to focus on German-language messages, I used the `langdetect` package `langdetect` to all messages. Results show that German content is clearly the predominant language with $\sim$ 83% (22.803 messages), followed by English ($\sim$ 3.8%, 3.823 messages), and then French, Italian, and Dutch with less than 1% each. Language detection was sometimes flawed for mixed-language texts. These were removed manually when encountered in the annotation process. After constraining the dataset to German-language texts, 22,803 messages remained, out of which texts with minimum length of 100 characters were selected, resulting in 14,605 remaining messages. Out of these, I selected twenty percent as a random sample, resulting in 4,021 messages. Since earlier findings in the context of our research project 'Digitaler Hass' revealed that Telegram channels often contain a rather high number of very similar messages, I removed such similar messages based on Levenshtein distance with a treshold of 0.8, resulting in 3,095 remaining messages.

While defining a minimum length for the sample seems reasonable, because the selected texts are likely to be comprehensible and contain a relevant number of entities, this constraint does not reflect the fact that social media texts are often of very short length. In fact, almost 50% of the messages were

[2]For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/blob/main/src/notebooks/preprocessing/v02_preprocessing.ipynb`.

[3]For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/data_selection`.

shorter than 100 characters. To create a realistic sample, I included some of these short texts as well, using the same procedure as described above, except for the sampling size which was 50% instead of 20%. After random sampling and removing similar messagess, 540 short messages remained and were added to the first sample.

6.1.4 Annotation

Data annotation turned out to be one the most challenging tasks of my project. First of all, existing annotation schemes for NER vary greatly in terms of the detailedness and precision of provided definition, the considered entity types, their granularity, and last but not least in how they handle social media-specific phenomena like user mentions, urls, or hashtags.[4]

I decided to create my own annotation scheme including entity type definitions, detailed instructions and a number examples to ensure consistent annotations also for ambiguous cases. The full annotation schema and guide can be found in the appendix (A). The tag set covers the six entity types person, location, organization, time, date and action and follows the BIO tagging scheme (see section ??). I used Labelstudio[5] for annotation. The O-tag was assigned automatically to all tokens left unlabeled.[6]

6.1.5 Telegram Corpus Statistics

Due to limited time and resources, I did not annotate the whole sample. The final annotated corpus consists of 500 documents including 2,509 annotated named entities. Table 6.1 shows the distribution of the different entity types after the train-test-validation split. Locations are by far the most often mentioned entity types (∼51%), followed by dates (∼17%), and action mentions

[4]An extremely tentative example is the annotation guide for the W-NUT 2015 shared task on NER in tweets: `https://docs.google.com/document/d/12hI-2A3vATMWRdsKkzDPHu5oT74_tG0-PPQ7VN0IRaw/edit`.

[5]https://labelstud.io/

[6]For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/blob/main/src/notebooks/4.2_training/01_prepare_datasets/prepare_telegram_annotation_results.ipynb`

(∼12%). Person names occur least frequently.

Table 6.1: Distribution of entities in Telegram corpus

	LOC	PER	ORG	DATE	TIME	ACTION	total
train	1.140	76	132	321	188	253	2.110
test	66	17	26	62	19	29	219
validation	73	17	18	40	9	23	180
total	1.279	110	176	423	216	305	2.509

6.2 The DFKI Smartdata Corpus

Compared to other NER datasets, the Telegram corpus is of very small of size. The GermEval 2014 corpus for example consists of 31,300 sentences containing more than 41.000 named entities (cf. Benikova et al. 2014. However, creating a larger corpus would have required more resources for annotation. I therefore extended my dataset by adapting the Smartdata corpus (Schiersch et al. 2018) for my task.

The Smartdata corpus consists of 2,598 German-language manually annotated documents which were sampled from online resources covering three genres: newswire texts, Twitter, and traffic reports via RSS feeds from radio stations, police and railway companies. The dataset contains a total of 22,075 named entities and is available under a permissive CC-BY 4.0 license (cf. ibid.).

Even though still relatively small-sized compared to standard NER datasets, the corpus is useful for my purpose for two reasons: The corpus is designed for the recognition of events and therefore considers time and date mentions as named entities. Second, it was created from online documents and is thus more similar to the Telegram corpus than standard newswire corpora for NER.

However, the tag set of the corpus is not readily combinable with the Telegram corpus, since the Smartdata corpus is based on a (partially) fine-grained annotation scheme, tagging organization mentions as organization or organization-company, and location mentions as either location, location-city, location-street, location-route, or location-stop. Furthermore, the dataset contains a number of entity types irrelevant for my task such as duration, distance, disaster type, or trigger events, while the entity type action from the Telegram corpus is missing. Last but not least, given the fact that the two corpora were annotated separately, annotation inconsistencies are likely to exist and negatively affect model training.

6.2.1 Label Mapping

Given the different tag sets, combining both datasets requires an adjustment of labels and annotations. I decided to use the tag set of the Telegram corpus as a reference, and adjust the Smartdata corpus accordingly.[7] I mapped all irrelevant entity tags (e.g. duration, disaster type, trigger event) to the `O` tag, and most of the fine-grained entity tags to their coarse-grained equivalent (e.g. organization-company to organization), provided that definition of the fine-grained entity was considered to sufficiently match the entity definition from the Telegram corpus. Table A.2 in the appendix provides a detailed overview of the label mapping.

6.2.2 Adjusted Smartdata Corpus Statistics

The adjusted Smartdata corpus contains a total of 14,557 entities, distributed as shown in table 6.2. We can see that as in the Telegram corpus, locations represent the largest class (∼47%), followed by organizations (∼27%), and dates (∼13%).

Table 6.2: Distribution of entities in Smartdata corpus after label mapping

	LOC	**PER**	**ORG**	**DATE**	**TIME**	**ACTION**	**total**
train	5.526	978	3.223	1.511	556	0	11.794
test	681	100	307	168	69	0	1.325
validation	688	127	387	167	69	0	1.438
total	6.895	1.205	3.917	1.846	694	0	14.557

[7]For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/blob/main/src/notebooks/4.2_training/01_prepare_datasets`.

6.3 The Final Corpus

Combining the Telegram corpus and the adjusted Smartdata corpus resulted in a total of 3,098 documents including 17,066 named entities with a class distribution of as shown in table 6.3.

Locations make up for almost half of all entities (∼48%) in the combined corpus, followed by organizations (∼24%), and date mentions (∼13%). Less frequently mentioned are persons (∼8%), and times (∼5%). Action mentions make up for less than 2%, resulting from the absence of this entity type in the Smartdata corpus. The dataset is thus heavily imbalanced which is known to negatively affect training.

Table 6.3: Distribution of entities in the final corpus

	LOC	**PER**	**ORG**	**DATE**	**TIME**	**ACTION**	**total**
train	6.666	1.054	3.355	1.832	744	253	13.904
test	747	117	333	230	88	29	1.544
validation	761	144	405	207	78	23	1.618
total	8.174	1.315	4.093	2.269	910	305	17.066

6.4 Models

I fine-tuned five different pretrained transformer models from the three different encoder model architectures BERT, DistilBERT, and XLM-R. The models differ not only in terms of their architecture, but also regarding size, sources, and languages of the corpora used for pre-training.

6.4.1 BERT Models

BERT models are language models pretrained on the tasks of masked language modeling and next sentence prediction (cf. Devlin et al. 2018). I selected two German-language BERT models for fine-tuning:

dbmdz/bert-base-german-uncased

This model is a German-language model pretrained on 16GB of data taken from Wikipedia, but also large online corpora created via web crawlers. I used the uncased version of the model, meaning that the model does not consider capitalization of words, since capitalization in user-generated text data is often unreliable (see section2.6). The cased version of the model was evaluated for NER fine-tuning on the CoNLL 2003 and GermEval2014 datasets, achieving F1 scores of 84.52 and 86.89.[8]

Model card: `https://huggingface.co/dbmdz/bert-base-german-uncased`

deepset/gbert-base

This German-language BERT model was pretrained on 163.4GB of data mainly consisting of scraped online resources (cf. Chan, Schweter, and Möller 2020). Fine-tuning for NER was evaluated on the GermEval2014 dataset, with an F1 score of 87.98 (cf. ibid.).

Model card: `https://huggingface.co/deepset/gbert-base`

6.4.2 DistilBERT Model

DistilBERT models are distilled version of BERT models, which allow faster training and require less memory, while achieving accuracy close to regular BERT models (cf. Tunstall et al. 2022, p. 80).

distilbert-base-multilingual-cased

This model is a distilled version of a multilingual BERT model, pretrained on a masked language modeling task using a concatenation of Wikipedia in 104 languages. It is reported to be twice as fast as its BERT counterpart.[9] The model is cased and thus considers capitalization. Evaluation results for NER fine-tuning are not provided.

Model card: `https://huggingface.co/distilbert-base-multilingual-cased`

[8] See `https://github.com/stefan-it/fine-tuned-berts-seq`, accessed on 2022/06/08.

[9] `https://huggingface.co/distilbert-base-multilingual-cased`, accessed on 2022/06/08.

6.4.3 XLM-Roberta Models

RoBERTa models modify the pretraining scheme of BERT models by dropping the task of next sentence prediction, and are trained longer and with more training data. This has shown to improve results compared to original BERT models (cf. Tunstall et al. 2022, p. 80). XLMs are cross-lingual language models. Their pretraining includes inputs from multiple languages, and extends the masked language modeling task to translation language modeling (cf. ibid., p. 80). XLM-Roberta models combine these two approaches by performing multilingual pretraining with large amounts of training data (cf. ibid., p. 80).

xlm-roberta-base

This model was pretrained on 2.5TB of webcrawled data covering more than 100 languages (cf. Conneau et al. 2019. It was evaluated for NER fine-tuning on the CoNLL 2002 and CoNLL 2003 datasets, achieving an F1-score of 84.60 for German when fine-tuned on each language set separately (cf. ibid.).

Model card: `https://huggingface.co/xlm-roberta-base`

cardiffnlp/twitter-xlm-roberta-base

The last model is an XLM-RoBERTa model pretrained on ∼198M tweets in over thirty languages (cf. Barbieri, Anke, and Camacho-Collados 2021), and thus the only model pretrained on social media texts. The evaluation of the model in the reference paper focuses on Twitter-specific tasks like e.g. emoji prediction, irony, hate speech and offensive language detection, or sentiment analysis. Evaluation of NER fine-tuning is not provided (cf. ibid.).

Model card: `https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base`

6.5 Experimental setup

In a first iteration, each model was fine-tuned on the Telegram corpus and the adjusted Smartdata corpus separately. The second iteration consisted of fine-tuning each model on the combined corpus. Before I report my results, I will outline the necessary steps for training and evaluation. The procedure was the same for all models to ensure comparability of results.

6.6 Defining a Baseline

Defining an appropriate baseline for my task is difficult due to a various reasons: The evaluation results for German-language NER fine-tuning for the chosen models are not directly comparable, since the standard datasets CoNLL2003 and GermEval2014 were used. These datasets differ greatly from my training corpus in terms of size, genre, and number of classes.

The shared tasks on social media NER described in **??** are roughly comparable to my setup in terms of corpus size, genre, and number of classes. However, all systems used in them represent outdated approaches to NER. This fundamentally limits the comparability of their results with my findings.

I decided to focus on state of the art results. Hence I will use the F1 score of the `deepset/gbert-base` model fine-tuned on the GermEval 2014 corpus. My baseline is thus F1=87.98.

6.7 Fine-tuning an NER model

The following implementation steps for NER fine-tuning were adapted from Tunstall et al. 2022, pp. 87–122.[10]

6.7.1 Preprocessing: Tokenization and Encoding

Tokenizers for transformer models takes care of several pre-processing steps including normalization, pretokenization, subword tokenization, and post-processing (cf. ibid., pp. 94–95). Transformer models rely on subword tokenization, combining character- und word-based tokenization by splitting up

[10]For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/4.2_training/02_training`.

words rarely occurring in the training data into smaller units and retaining frequent words as a whole. This allows to handle complex words and misspellings, but at the same time keep the input length at a processable size (cf. Tunstall et al. 2022, p. 33). During pretraining, different algorithms such as WordPiece, Byte-Pair Encoding (BPE), or SentencePiece are applied to the training data, allowing the model to learn which words should be split up further (cf. HuggingFace 2022c). Tokenizers for my experiments were loaded via the `from_pretrained()` method of the `AutoTokenizer` class.

Label alignment for tokenized inputs

While the tokenizer splits up the input strings into subwords, the named entity tags in the training corpus are usually assigned to entire words. Therefore we need to take care of aligning the tags with their corresponding subtokens.

Every word in the input is assigned a unique `word_id`. This `word_id` also assigned to all corresponding subtokens. We can therefore use this id for label alignment. Subword tokenization also needs to be considered for later evaluation, since the loss function of the model needs to know how to handle subtokens. My implementation follows Tunstall et al. and only considers the first subtoken of a word for calculating the loss. This can be implemented by assigning the value -100 as `label_id` to all subtokens which have the same `word_id` as their predecessor. This tells the cross entropy loss function to ignore these subtokens, hence only the model's prediction for the first subtoken of a word will be considered (cf. Tunstall et al. 2022, p. 104). I used `is_split_into_words=True` to tell the tokenizer that the input data is already tokenized. Shorter inputs were padded, longer inputs were truncated to the maximum length of the respective model.

After preprocessing and encoding, the dataset includes numerical `input_ids`, representing the input tokens, and numerical `labels`, representing the corresponding named entity tags, as shown in figure 6.2. The tokenizer has also returned an `attention_mask` for each input document, telling which token should be attended and which not (cf. HuggingFace 2022a). Furthermore, the tokenizer created `token_type_ids`. These are not relevant for my task, but used for tasks in which an input contains two sequences, e.g. for question answering or tasks involving sentence pairs such as next sentence prediction. In these cases, the `token_type_ids` tell the model to which of the two sequences a token belongs (cf. HuggingFace 2022b). We now have encoded our textual input, consisting of tokens and labels, into a numerical input which can be processed by the model.

	input_ids	token_type_ids	attention_mask	labels
0	[102, 20739, 8497, 30914, 10412, 30925, 22171,...	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...	[-100, 0, -100, -100, -100, -100, -100, -100, ...
1	[102, 17329, 14705, 18315, 17329, 19281, 30904...	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...	[-100, 0, -100, -100, 5, -100, -100, -100, -10...
2	[102, 718, 1412, 6556, 17077, 3859, 136, 8139,...	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...	[-100, 0, 3, -100, -100, -100, 0, 3, -100, 0, ...
3	[102, 493, 566, 1483, 1073, 818, 3035, 853, 32...	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...	[-100, 1, 2, 2, 2, 0, 9, -100, -100, 10, 5, 0,...
4	[102, 15591, 17329, 7938, 2032, 305, 1055, 853...	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...	[-100, 0, 0, -100, -100, -100, -100, 0, 0, 0, ...

Figure 6.2: Training data after tokenization and encoding

6.7.2 Loading and Configuring a Model

As a next step, we need to load the model and adjust its configuration to make it fit our dataset. We do this by loading the `AutoConfig` of the pretrained model and handing over the number of labels (`num_labels`)in our dataset as well as the mapping between numerical and textual labels, defined as (`id2label`) and (`label2id`). Since NER fine-tuning is about classification of single tokens for NER, we choose a model for token classification.

6.7.3 Defining Training Arguments

The following training arguments were defined for all models and datasets (some arguments for logging and saving the model are omitted):

```
from transformers import EarlyStoppingCallback

BATCH_SIZE = 2
NUM_EPOCHS = 10
WEIGHT_DECAY = 0.01
CALLBACK = EarlyStoppingCallback(early_stopping_patience=3)
LOAD_BEST_MODEL_AT_END = True,
METRIC_FOR_BEST_MODEL='eval_loss'
```

By `NUM_EPOCHS` we define how many times the model will work through the whole dataset. `BATCH_SIZE` defines the number of examples processed per device (e.g. a GPU) in each training step. A batch size of 2 is admittedly small, but was chosen to avoid running into memory errors during training.

`WEIGHT_DECAY` is used as one out of many regularization technique to prevent overfitting. I also defined a `CALLBACK` to implement early stopping during training. Callbacks can be used to customize the training behaviour of transformer models and take decisions during training. Here, the decision is to stop if the provided `METRIC_FOR_BEST_MODEL` deteriorates the third time in a row, defined via `early_stopping_patience=3`. I used the model's loss on the validation set as reference metric for early stopping and loading the best model at the end of the training process.

Data Collator

I also used a data collator which takes care of padding for labels and inputs. Since padding was already carried out during tokenization and encoding, this can be understood as an alternative implementation.

6.7.4 Evaluation Metrics

For evaluation purposes, I defined a method `compute_metrics` which returns the F1 score, precision, recall, and accuracy per training epoch. Analogue to the process of label alignment in section 6.7.1, the model's predictions are aligned using the `label_id` -100 for subtokens to be ignored by the loss function.

6.7.5 Training

For training, I initialized a `Trainer` object and passed the loaded model, the training arguments, the tokenizer, the data collator, the callback, and the encoded data splits for training and validation to it.

6.8 Evaluation Results

The following section explores the results of fine-tuning the selected models on the two datasets separately, and on the final combined dataset. Recall the baseline is defined as F1=87.98.

Generally we can observe that for all experiments, training was stopped after a maximum of three epochs, in some cases already after one epoch, indicating that the validation loss starts to increase rather soon in all experiments. This is not surprising given the rather small size of the provided data sets, which easily leads overfitting.

The defined baseline could not be reached or outperformed in any of my experiments. Regarding the separate experimental setups, it can be observed that fine-tuning exclusively using the Telegram corpus overall yields the lowest results: None of the F1 scores increases 80.0 and are thus far from the baseline. The DistilBERT model achieves the lowest score with F1=64.81, the German-language *deepset/gbert-base* model performs best with F1=74.82.

Table 6.4: Fine-tuning different models on Telegram corpus

Model	**F1 score**	**Validation Loss**	**Epoch**
`distilbert-base-multilingual-cased`	64.81	0.2910	1
`dbmdz/bert-base-german-uncased`	74.40	0.2285	2
`deepset/gbert-base`	**74.82**	0.2524	2
`xlm-roberta-base`	73.30	**0.2459**	3
`cardiffnlp/twitter-xlm-roberta-base`	71.03	0.2648	3

Fine-tuning using only the adjusted Smartdata corpus yields the most promising results, as shown in table 6.5.

Overall, the models perform better with most F1-scores exceeding F1 of 80.0

and therefore getting closer to the baseline. An exception is the dmbdz/bert-base-german-uncased model achieving F1=61.56. Once again, the *deepset/gbert-base* model performs best with F1=85.35 which is only ~0.2 below the baseline, which is promising especially when considering that the baseline was achieved by fine-tuning on a much larger corpus.

Table 6.5: Fine-tuning different models on adjusted Smartdata corpus

Model	**F1 score**	**Validation Loss**	**Epoch**
`distilbert-base-multilingual-cased`	82.03	0.1456	2
`dbmdz/bert-base-german-uncased`	61.56	0.2560	3
`deepset/gbert-base`	**85.35**	**0.1114**	3
`xlm-roberta-base`	82.64	0.1445	1
`cardiffnlp/twitter-xlm-roberta-base`	82.66	0.1344	3

The results for fine-tuning on the combined corpus are shown in table 6.6. We can observe that the scores improved compared to fine-tuning only on the Telegram corpus, which is plausible considering the larger size of provided data. However, we notice a decrease of scores compared to fine-tuning only on the adjusted Smartdata corpus. One explanation for this could be annotation inconsistencies between the two corpora, or the imbalanced class distribution, especially the low number of examples for the entity type action.

Table 6.6: Fine-tuning different models on combined corpus

Model	**F1 score**	**Validation Loss**	**Epoch**
`distilbert-base-multilingual-cased`	66.34	0.2543	2
`dbmdz/bert-base-german-uncased`	77.75	0.1751	1
`deepset/gbert-base`	79.22	**0.1751**	2
`xlm-roberta-base`	**79.88**	0.1842	3
`cardiffnlp/twitter-xlm-roberta-base`	77.68	0.2061	3

Judging only by the F1-score, the best model on the combined corpus was the *xlm-roberta-base* model with F1=79.88, followed closely by the *deepset/gbert-base* model with F1=79.22. Regarding the validation loss, the latter performs better than the *xlm-roberta-base* model. Furthermore, the *deepset/gbert-base* model achieves a better precision, as shown in 6.7, indicating that its results can be considered more relevant than those of the *xlm-roberta-base* model. Against this background, I selected the *deepset/gbert-base* model as preliminary best model for further error analyses.

Table 6.7: Detailed metrics for two best models

Model	**F1 score**	**Precision**	**Recall**	**Validation Loss**	**Epoch**
`deepset/gbert-base`	79.22	**78.16**	80.32	**0.1751**	2
`xlm-roberta-base`	**79.88**	76.82	**83.19**	0.1842	3

6.9 Error Analysis

With an F1 score of 79.22, my preliminary best model lags behind the baseline of F1=87.98 by 8.76. Even when considering the smaller training corpus size, the performance of my model cannot be considered state of the art performance for NER. Furthermore, even though the combined corpus is of larger size than the Smartdata corpus, results deteriorate when fine-tuning on the combined corpus. These findings demand for an error analysis to identify potential improvements for future experiments.

Token-level Errors

Figure 6.3 shows the tokens with the highest total loss on the validation set of the combined corpus for the *deepset/gbert-base* model.

The token `.` has the highest total loss of 118.7. This is not surprising, considering its frequent occurence (count=647) in the validation set. We can see that its mean loss of 0.18 is rather low compared to other listed tokens. The fact that the token `.` is often part of named entities of type date and time, but also often is not a named entity, but a regular character, might also confuse the model. In general we see that a lot of special characters are among the tokens with the highest total loss. It can be assumed that this is partly due to their occurrence inside of named entities. Furthermore, the words `Uhr`, `Protest`, and `Kund` appear. The word `Uhr` can mostly assumed to occur as part of time mentions. Both annotation schemes include the word as part of time named entities, therefore I cannot be assumed that annotation inconsistencies are the main reason for its high total loss. The tokens `Protest`, and `Kund` are part of named entities of type action, which is underrepresented in the dataset, making it plausible that they are often misclassified by the model. They also have by far the highest mean loss of all listed tokens.

	0	1	2	3	4	5	6	7	8	9
input_tokens	.	-	#	,	@	Uhr	Protest	Kund	(	:
count	647	351	110	402	150	64	6	4	112	262
mean	0.18	0.27	0.77	0.12	0.3	0.65	6.44	9.49	0.29	0.12
sum	118.7	94.66	84.68	46.93	45.02	41.5	38.62	37.95	32.45	30.55

Figure 6.3: *deepset/gbert-base*: Tokens with highest total loss

Class-level Errors

For a more general level error analysis, I also explored the loss per label on the validation set. Figure 6.4 shows the results, classes are ranked by highest mean loss. The labels `I-ACTION` and `B-ACTION` have by far the highest mean loss with 12.17 and 11.95. Their mean loss exceeds those of other classes by a factor of up to 28.5. These findings clearly indicate that the fine-tuning was negatively affected by the underrepresented class action in the training data. Further fine-tuning thus requires handling this problem to allow for an improve of performance. However, the remaining named entity labels are fairly balanced regarding their mean loss, with an upper tier between 1.02 and 0.7 including `B-DATE`, `I-ORG`, `I-LOC`, `I-TIME`, `B-ORG`, and a lower tier between 0.63 and 0.43 including `B-TIME`, `I-DATE`, `B-LOC`, `B-PER`, and `I-PER`. The `O` class has by far the lowest mean loss which is not surprising giving the high number of examples for it in the training data. Once again, we can see that the action class causes the most errors, with `B-ACTION` being predicted as `O` in almost all cases, and `I-ACTION` in all cases.

	0	1	2	3	4	5	6	7	8	9	10	11	12
labels	I-ACTION	B-ACTION	B-DATE	I-ORG	I-LOC	I-TIME	B-ORG	B-TIME	I-DATE	B-LOC	B-PER	I-PER	O
count	6	47	214	251	315	61	416	95	166	852	142	127	12862
mean	12.27	11.95	1.02	0.99	0.92	0.91	0.7	0.63	0.55	0.47	0.45	0.43	0.08
sum	73.64	561.81	217.98	248.98	288.9	55.59	291.28	60.11	91.22	403.99	64.46	54.3	988.29

Figure 6.4: *deepset/gbert-base*: Labels ranked by mean loss

Confusion Matrix

The confusion matrix of the model as shown in figure 6.5 is mainly diagonal, indicating that the majority of predicted labels corresponds with the true label for most of the classes. Not surprisingly, the majority class `O` tag is most commonly predicted in case of wrong predictions. Once again, we can observe that the action class causes by far the most errors, with tokens assigned a `B-ACTION` label predicted as `O` in most of the cases, an tokens assigned a `I-ACTION` label in all cases.

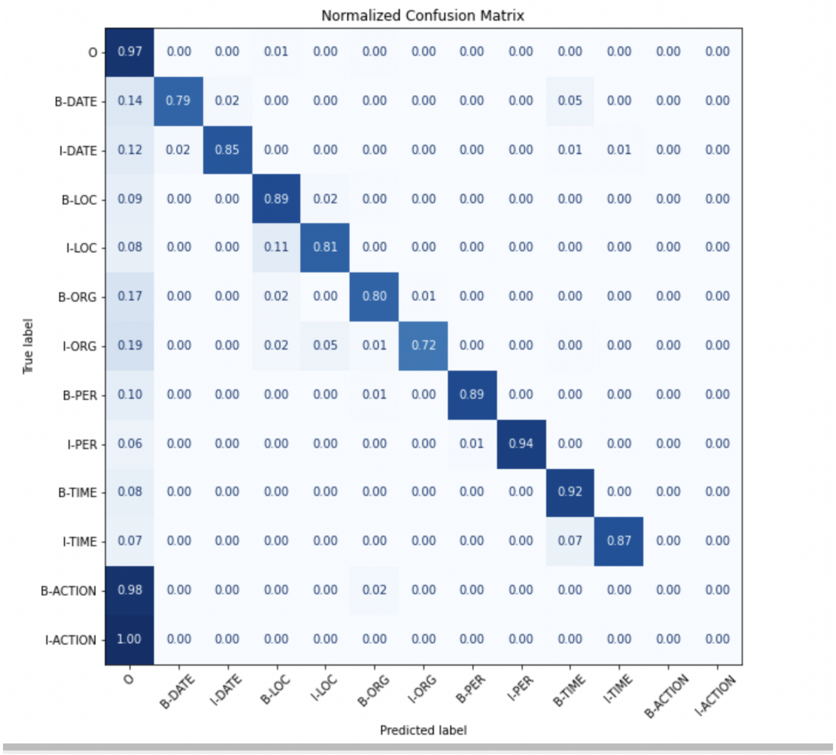


Figure 6.5: *deepset/gbert-base*: Confusion matrix

6.10 Application of the Custom Model

For the application of my custom model, I used Hugging Face's `pipeline()` to extract named entities from the ten example messages used already in chapter 5, and compared the results of the custom model to those of the German-language Flair model. I furthermore extracted named entities from a larger sample of messages from the channel 'Demotermine'.[11]

Since the pipeline returns subtokens, some post-processing of the results was required in order to obtain full words, and to carry out label alignment. The corresponding B- and I-tags were mapped to the general labels `PER`, `LOC`, `ORG`, `TIME`, and `DATE`. I also experimented with parsing date and time mentions using the `ctparse` package.[12]

I would like to point out that this part of my implementation is still work in progress, both regarding the quality of the model and the post-processing steps, especially the parsing of times and dates.

6.10.1 Example Messages

Table 6.8 shows the results of the German-language Flair model from chapter 5 compared to the results of the custom model. We can observe that the custom model successfully detects the non-standard named entity types `DATE` and `TIME` in messages no. 2, 3, 4, 6, 8, and 10, while the Flair model does not support them and hence cannot recognize them. Furthermore, we can see in message no. 3 that the custom model has learned to recognize non-standard location mentions such as 'Ausfahrt' or 'Berliner Speckgürtel'. This is useful for a more fine-grained monitoring of protest mobilization, and also helps to extract rather informal mobilization information. We can also observe some errors by the model, e.g. the postal code in message no. 3 is correctly recognized as location, which is an improvement compared to the Flair model, but the result is incomplete, missing the last digit. The model also misses

[11] For implementation details, see `https://github.com/eli-quereli/ner-german-telegram/tree/main/src/notebooks/4.2_training/03_results`.

[12] `https://ctparse.readthedocs.io/en/latest/`

the action mention 'Demo-Berlin', which is not surprising given the poor performance of the model for this class.

6.10.2 Larger Sample

3,634 messages initially selected as sample for annotation includes the 500 annotated messages of which 80% were used as training data for fine-tuning the custom model.

applied custom model using the pipeline, evaluate the most common named entities per entity type

dates and times: compare initial results versus results after parsing

6.11 Summarization

In this chapter I described my efforts to train a custom NER model for extracting named entities of type person, location, organization, date, time, and action from German-language Telegram messages. I describe my efforts of building a Telegram corpus by selecting sample messages and annotating them using a newly developed annotation scheme. I used additional training data from the Smartdata corpus, adjusting its tag set to make it fit the Telegram corpus. I ran a number of experiments to evaluate the model of five different pretrained transformer models, first on each corpus separately, and then on the combined corpus. For the preliminary best model, a fine-tuned version of the German-language BERT model *deepset/gbert-base*, I conducted further error analysis and evaluation on token and class level. Finally, I apply this model to the ten example messages used in chapter 5, and to a larger sample of Telegram messages.

My findings confirm the potentials of using transformer models for NER fine-tuning for building a custom model for NER in social messengers channels. The preliminary best model achieves F1 score of 79.22. Even though it lags behind the baseline of 87.98, its performance for the standard named entity types location, person, and organization, and for the non-standard

Table 6.8: Detected entities for German-language example messages: Flair vs. custom model

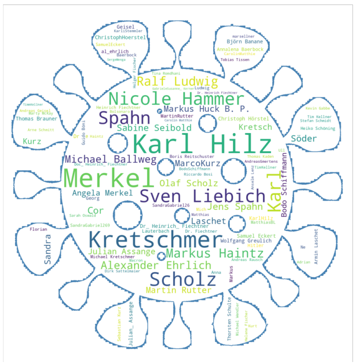
No.	Text	flair/ner-german	gbert-base/finetuned
1	Freiburg Aufzug mit Endkundgebung am Platz der alten Synagoge . Live Musik ! Details 19062021 Raus auf die Straße	Freiburg[LOC]	Freiburg[LOC]
2	GEHT DEMONSTRIEREN ! - Message von Elijah Tee an die Dresdner & anderen Demonstranten in Deutschland Morgen am 13.06. um 16.30 Uhr und bundesweit überall , wo es Mitdenker gibt . - Bilder und Videos hochladen - Diskussion Fahren oder Mitfahren	Elijah Tee[PER], Deutschland[LOC]	Elijah Tee[PER], Dresdner[LOC], Deutschland[LOC], 13.06.[DATE], 16.30 Uhr[TIME]
3	15831 Blankenfelde Mahlow Berliner Speckgürtel Ausfahrt Lichtenrade am 14.03.2021 14032021 Raus auf die Straße	Blankenfelde Mahlow[LOC], Lichtenrade[LOC]	1583[LOC], Blankenfelde[LOC], Mahlow[LOC], Berliner Speckgürtel[LOC], Ausfahrt[LOC], Lichtenrade[LOC], 14.03.2021[DATE]
4	2021 10 11 Sven Liebich fragt Ministerpräsident Haseloff nach dessen Verständnis von Versammlungsfre - Sven Liebich 0:23 Raus auf die Straßen Übersicht / Overview	Sven Liebich[PER, Haseloff[PER], Sven Liebich[PER]	20211011[DATE], Sven Liebich[PER], Haseloff[PER], Sven Liebich[PER], 0[TIME]
5	FÜR ALLE DEMOKRATEN GEFAHR : Die Polizei blockiert den Stern ! Deswegen laufen die Menschen vom Ernst-Reuther-Platz nach Südosten zum Breitscheitplatz und dort weiter auf den Kudamm ! ! Schließt euch an : Neue Kantstraße direkt Richtung Osten auf den Breitscheitplatz ! Von der Straße des 17. Juni Richtung Süden zum Kudamm !	FU[ORG], Ernst-Reuther-Platz[LOC], Breitscheitplatz[LOC], Kudamm[LOC], Kantstraße[LOC], Breitscheitplatz[LOC], Kudamm[LOC]	Ernst-Reuther-Platz[LOC], Breitscheitplatz[LOC], Kudamm[LOC], Neue Kantstraße[LOC], Breitscheitplatz[LOC], 17.Juni[DATE], Kudamm[LOC]
6	Dr. Gunter Frank : Wie Moralismus dazu führt das man Feinde sieht und vernichten will — Antihygienista - Offene Gesellschaft Kurpfalz 15:04 Raus auf die Straßen Übersicht / Overview	Gunter Frank[PER], Antihygienista[ORG], Offene Gesellschaft Kurpfalz[ORG]	Gunter Frank[PER], Antigienista- Offene Gesellschaft[ORG], Kurpfalz[LOC], 15:04[TIME]
7	Einfach nur Krank solche Politiker ! Was sollen wir mit ihm machen wenn unsere Kinder anhand dieser Impfung nen schaden erleiden ? Wer ist dann dafür verantwortlich , wen schmeißen wir dann in den Knast und nehmen sein ganzes Geld weg , sollen wir das mit dir machen ? Raus auf die Straße	*no entities detected*	*no entities detected*
8	27.11.2021 Frankfurt Durchsagen von gemischt mit Musik . Super Idee Für mehr Berichte folgen auf ... Raus auf die Straßen Übersicht / Overview	Frankfurt[LOC]	27.11.2021[DATE], Frankfurt[LOC]
9	Demo-Berlin : Der Mann mit dem Schild wurde soeben von der Polizei festgenommen . Circa 7 Einsatzwägen sind vor Ort und umzäunen den Park großräumig . Am Humboldthain gibt es nichts mehr zu gewinnen . # b2908 Raus auf die Straßen	Demo-Berlin[LOC], Humboldthain[LOC]	Berlin[LOC], Humboldthain[LOC]
10	Anthony Fauci : Definition von " vollständig geimpft " könnte geändert werden — Epoch Times Nachrichten 6:59 Raus auf die Straßen Übersicht / Overview	Anthony Fauci[PER], Epoch Times Nachrichten[ORG]	Anthony Fauci[PER], Epoch Times Nachrichten[ORG], 6:59[TIME]

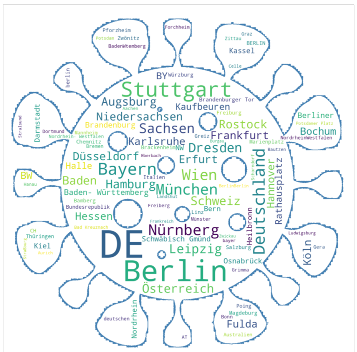
Table 6.9: Application of custom model: Detected named entities in larger sample of Telegram messages

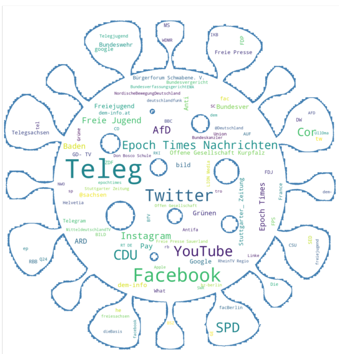
LOC	**PER**	**ORG**	**DATE**	**TIME**	**ACTION**
13,656	1,513	1,926	2,453	3,973	0

entity types time and date is promising with most of the related classes achieving F1 scores above 0.8, some even above 0.9. However my results also indicate that fine-tuning for non-standard entity types like the class action is a more challenging task. My preliminary best model almost completely fails to detect entities of this type. It can be assumed that the model's poor performance for this class results from its underrepresentation in the training data. The use of such non-standard entity types thus requires more resources for annotating corpora, or should generally be reconsidered. Nonetheless, my model represents an improvement over the existing solutions explored in the previous chapter, since its application demonstrates that it allows for the detection of date and time mentions, representing non-standard entity types not supported by most existing NER models.

Table 6.10: Application of custom model: Detected named entities in larger sample of Telegram messages







Chapter 7

Conclusion

The aim of my thesis was to explore the potentials and limitations of NER for analyzing and monitoring online mobilization for protests in the context of the COVID-19 pandemic in Germany. The focus was on German-language messages from Telegram channels. I chose to extend the standard named entity types person, location, and organization by the three additional types date, time, and action since I consider them as particularly relevant for my concern.

7.1 Summarization

As a theoretical and conceptual basis, I provided an introduction to the task of NER (chapter 2) and present prior research on both German-language NER in noisy user-generated texts from social media, discussing the general challenges for NER and the specific challenges for NER in social media texts (chapter 3). My findings show that most existing approaches to NER focus on the standard entity types person, location, organization, and other/miscellaneous. Furthermore, like for many other NLP tasks, there is a focus on English-language data. While there are standard NER datasets for German-language texts, these most commonly consist of newswire data. NER corpora consisting of more informal texts such as tweets are commonly restricted to

English-language texts.

I then turned to state of the art approaches for NER: fine-tuning of transformer models, and provided a high level description of the key concepts of transformer models (chapter 4).

In a next step, I explored the applicability of existing NER models for my task by qualitatively analyzing their results in ten example messages from a Telegram channel (chapter 5). My qualitative analysis confirms that existing NER solutions widely unsuitable for my concern. The tested German-language Flair model yields promising results, but is restricted to the standard entity types and thus does not allow to extract date, time, or action mentions.

Finding that none of the analyzed models covers the named entity types required for my task, I then turned to building a custom NER model using fine-tuning of transformer models (chapter 6). For this, I created a manually annotated Telegram corpus, and extended it by an adjusted version of the Smartdata corpus. I evaluated a selection of five pre-trained transformer models, including both German-language and multilingual ones, by fine-tuning them on the Telegram and Smartdata corpus separately, and on a combined corpus. I evaluate my models using their F1 score as standard metric for NER. My preliminary best model achieves an F1 score of 79.22 on the combined corpus. I conducted further evaluation and error analysis for this model, focussing on token and class level errors. Finally, I applied my custom model to the previously used example messages, and compare its results to those of the Flair model analyzed in chapter 5. I also apply the model to a larger sample to demonstrate its possible use for automatically analyzing data from Telegram channels. My results show that building a custom NER models covering non-standard entity types is a challenging task. While the model achieves moderate to good performance for the standard entity types person, location, and organization, and also can be fine-tuned successfully to detect date and time mentions, it fails at recognizing entities of type action. This failure is most probably due to the underrepresentation of this class. Annotations for this class only exists in the smaller Telegram corpus, which results in heavily imbalanced training data. However, regarding the other classes, my experiments confirm that transformer models can be successfully fine-tuned to specific NLP tasks such as NER, even when only rather small amounts of labeled data are provided. With my thesis, I contribute to research about German-language NER in informal online domains such as messenger services, and provide first insights into the applicability of NER for monitoring efforts of journalists and non-governmental organizations. I furthermore developed a comprehensive annotation guide as a basis for future annotation efforts of Telegram messages and similar content.

7.2 Discussion

My results confirm that fine-tuning of transformer models is a promising approach for building NER applications covering non-standard entity types in informal text genres. Most of the fine-tuned models significantly outperform systems using classic machine learning based approaches presented at past shared tasks on NER in user-generated texts. However, none of them achieves state of the art results for the standard 4-class NER task. These findings demand for further discussion.

First and foremost, I assume that more data is required to obtain state of the art results for my task. The custom Telegram corpus consists of no more than 500 messages. Extending it by the Smartdata corpus resulted in a total of 3,098 documents, which is admittedly still rather small, especially when compared to the standard NER datasets commonly used for training and evaluating state of the art NER models. Furthermore, the entity type action is heavily underrepresented in my corpus since it was only annotated in the Telegram subset. The Smartdata corpus does not offer a corresponding label. I assume that this is among the main reasons for the poor performance of the model for this class. Not only is the model provided to few examples, but possible mentions of type action will be labeled as non-entity in the Smartdata corpus, which represents by far the larger proportion of the training data.

Another factor which might have negatively affected training might be the differences between the data included in the two corpora: While they both cover texts from online resources including mentions of events, they differ in terms of style and content. The fact that scores of the models for the combined corpus decreased compared to their performance on the Smartdata corpus only might indicate that the two corpora are too different for merging. While mixed-domain corpora can be of advantage for building flexible models, in this case the two subsets might also simply be too imbalanced in terms of size. Obtaining a larger sample of annotated Telegram messages could encounter this imbalance. Furthermore, the exclusion of mixed-language messages from the training data could be questioned. Since such mixed-language messages represent realistic social media data from messenger services, it could be argued that they should be included in the training data.

After conducting an error analysis of my preliminary best model, I would also reconsider my annotation scheme and the pre-processing for annotation. For example, the high error on special characters might indicate that they should not be included in entity spans, and that the annotation should be updated accordingly. Handling of hashtags should also be reconsidered: While I tried to include date and location mentions encoded in hashtags (e.g. #b2908), this might confuse the model. I would thus recommend to not include hashtags as named entities, but rather apply separate methods for extracting information from them.

7.3 Future Work

My results obtained by the preliminary best model give a first impression which kind of information can be extracted from Telegram messages. However, using this information for proper analyses and monitoring would require further post-processing. In this context, parsing of date and time information would be one important step. Representations of dates and times cover a great variety of styles in user-generated texts, making this subtask more challenging than for more formal texts e.g. from news articles. Furthermore, entity disambiguation should be carried out in order to map different representations of an entity to the same key. Once again, the informal nature of user-generated texts, including incomplete mentions and a wide variety of styles, e.g. regarding capitalization, should be considered for this task. The extraction of relationships between extracted entities would be a valuable contribution. The mere occurrence of e.g. a date is only of limited use for monitoring mobilization efforts. being able to relate such a date mention to a location and time might be of crucial importance for monitoring purposes. Regarding the use of NER in production, implementing a search function for the obtained results, as well as improved visualization, e.g. as a map containing locations, associated with date, time or person and organization mentions would be further valuable contributions.

I generally consider efforts to make NLP approaches such as NER available for journalists and non-governmental an important and beneficial task. Given the ongoing successful online mobilization strategies of far-right activists in Germany, tools for automated the analysis of large amounts of text data from online platforms is urgently needed and should be made available to those who are working day by day to raise awareness on the violent potential in these networks.

Glossary

Information Extraction The task of extracting information from unstructured or semi-structured machine-readable data, e.g. texts.. 6

Named Entity Recognition The task of extracting named entities from text.. i

Natural Language Processing Natural Language Processing combines approaches from computer science, linguistics, and artificial intelligence. The main focus is to build systems and computational techniques for the automated processing and analysis of human language.. 4

Acronyms

ACE Automatic Content Extraction. 6

BiLSTM Bidirectional Long Short Term Memory Network. 27

CoNLL Conference on Computational Natural Language Learning. 15, 18, 27, 30, 41–43

CRF Conditional Random Field. 20, 27

DARPA Defense Advanced Research Projects Agency. 6

IE Information Extraction. 6, *Glossary:* Information Extraction

LSTM Long Short Term Memory Network. 20

MLM Masked Language Modeling. 24

MUC Message Understanding Conference. 6

NER Named Entity Recognition. i, 4, 6–13, 15–19, 24–28, 30, 31, 33, 34, 37, 38, 41–43, 45, 49, 53, 55, 57–59, 61, *Glossary:* Named Entity Recognition

NLP Natural Language Processing. 4, 6, 8, 11, 12, 18, 19, 26, 27, 57, 59, 61, *Glossary:* Natural Language Processing

NSP Next Sentence Prediction. 24

OOV out of vocabulary. 13

RNN Recurrent Neural Network. 20

TAC KBP Text Analysis Conference Knowledge Base Population. 6

W-NUT Workshop on Noisy User-generated Text. 17, 18, 37

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Appendix

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Appendix A

Additional Material

Table A.1: Overview of entity types of the English spaCy model

Entity type	**Definition**
CARDINAL	Numerals that do not fall under another type.
DATE	Absolute or relative dates or periods.
EVENT	Named hurricanes, battles, wars, sports events, etc.
FAC	Buildings, airports, highways, bridges, etc.
GPE	Countries, cities, states.
LANGUAGE	Any named language.
LAW	Named documents made into laws.
LOC	Non-GPE locations, mountain ranges, bodies of water.
MONEY	Monetary values, including unit.
NORP	Nationalities or religious or political groups.
ORDINAL	"first", "second", etc.
ORG	Companies, agencies, institutions, etc.
PERCENT	Percentage, including "%".
PERSON	People, including fictional.
PRODUCT	Objects, vehicles, foods, etc. (not services).
QUANTITY	Measurements, as of weight or distance.
TIME	Times smaller than a day
WORK_OF_ART	Titles of books, songs, etc.

Table A.2: Mapping of label schemas

Telegram Label	**Original Smartdata Label**
B-DATE	B-DATE
I-DATE	I-DATE
B-LOC	B-LOCATION, B-LOCATION_CITY, B-LOCATION_STOP, B-LOCATION_STREET
I-LOC	I-LOCATION, I-LOCATION_CITY, I-LOCATION_STOP, I-LOCATION_STREET
B-ORG	B-ORGANIZATION, B-ORGANIZATION_COMPANY
I-ORG	I-ORGANIZATION, I-ORGANIZATION_COMPANY
B-PER	B-PERSON
I-PER	I-PERSON
B-TIME	B-TIME
I-TIME	I-TIME
O	B-DISASTER_TYPE, I-DISASTER_TYPE, B-DISTANCE, I-DISTANCE, B-DURATION, I-DURATION, B-LOCATION_ROUTE, I-LOCATION_ROUTE, B-NUMBER, I-NUMBER, B-ORG_POSITION, I-ORG_POSITION, B-TRIGGER, I-TRIGGER

Annotation Guidelines

The following annotation guidelines are a combined adaptation of Benikova, Biemann, and Reznicek 2014, LDC 2005, Chinchor 1997, and Schön et al. 2018.

Named Entity - Basic Definition

Names are denominators for a single unique object in the real world. Only full noun phrases are considered as named entities. Pronouns and all other phrases can be ignored. Phrases only partly containing names (e.g. ‘deutschlandweit), or adjectives referring to entities (‘euklidisch’) can be ignored.

Only one entity type per token should be annotated. In case of multiple options (e.g. ‘Charité’ could be referred to as organization or as location, same for ‘Landtag’ or ‘Bundestag’), the context of usage should be considered for the final classification.

Named entities can consist of more than one token. The inital token of an entity should be tagged with the B-label of the corresponding entity type. The following tokens belonging to the same entity should be labelled with the I-label of the same type. For

Combined named entities including different entity types (e.g. ‘Wien-Demo’) should be tagged separately if possible (Wien[LOC]-Demo[ACTION]), or else tagged as the dominant entity type (here: [ACTION]) to avoid nested entities.

Whitespace should not be labelled as part of an entity. Token separators such aus /, - or . should be included if they are inside an entity sequence, and excluded if they separate two different entities, or an entity and a non-entity token. Hashtags should be included in the labeled span.

Dates and times can be represented by numbers (31.10.2019, 16:00), and strings ('Morgen', 'Uhr').

Emoticons, user mentions and (parts of) urls should not be tagged as named entities.

Definition of Different Entity Types

For annotation, the the 6 entity types PERSON, LOCATION, ORGANIZATION, DATE, TIME, and ACTION are used. The label set consists of B-tag and I-tag for each entity type, resulting in a total of 12 labels.

Table A.3 shows the core definitions and examples for the annotated entity types.

Specifications

Handling Mixed-language Messages

Skip mixed-language in which the actual message is non-German language.

Examples:

- *We are angry and pissed off - Raus auf die Straße Demotermine!*
- *Brilliant numbers in Carlisle ! Raus auf die Straßen @Demotermine*
- *MASSIMA DIFFUSIONE http://t.me/stopdittatura/1930 Raus auf die Straßen @Demotermine*

Table A.3: Definition of entity types for annotation guide

Label	**Description**	**Examples**
PER	Named, single individual, limited to humans. Fictional persons can be included, as long as they're referred to by name.	Karl Lauterbach, Merkel, Harry Potter, Kretscher, Karl Hilz, Markus
LOC	Names of politically or geographically defined locations, including countries, provinces, regions, cities, districts, addresses including streets, squares, and postal codes, (named) buildings or other man-made structures, shopping centers, parking lots, or restaurants, cardinal points. No metaphorical locations.	Deutschland, Frankreich, #BW, Berlin, Bundestag, Restaurant Walliserkanne, Straße des 17. Juni, Neptunbrunnen, Alex, Nollendorfplatz, Norden, Süden, Hellersdorf
ORG	A named corporate, governmental, or other organizational entity: Organization entities can be corporations, agencies, institutions, newspapers, tv/radio channels, other media platforms, political parties, non-governmental organizations, and political groups with a formal or informal organizational structure.	ARD, Polizei Berlin, Youtube, Telegram, Amadeu Antonio Stiftung, Querdenken711, Antifa, Bundesregierung, Gesundheitsamt, Französische Polizei, Facebook, Google, Bill Gates Foundation, KSV LiLi, FC St. Pauli, SGD-Ultras, Bündnis21 Hessen, Polizei, Polizeipräsidium Pforzheim
DATE	Absolute or relative date expressions	23.09.2022, 20210930, 3108, Morgen, Heute, 2019, #Sa2011
TIME	Specific, absolute time expressions referring to times of day, and time codes.	13:10, 13:12 Uhr, 16 Uhr, 6:49
ACTION	A specific instance of an action or event undertaken to express political beliefs, e.g. demonstrations, vigils, motorcades, walks in public.	Kundgebung, Demo, Demonstration, Spaziergang, Aufzug, Aufzug RufderTrommeln

Handling Incomprehensible Messages

Skip incomprehensible messages.

Example: *"Sterne-Restaurant Walliserkanne . Naudreck . Bersets faule Tricks . Uniarzt panikfrei . Herzmuskel . Neuer Beitrag #Walliserkanne Raus auf die Straßen Demotermine! Übersicht / Overview"*

Handling Short Messages

Skip short messages which do not include at least two named entities.

Examples:

- *"AnwaelteFuerAufklaerung Demotermine"*
- *"Egal wo ihr diesen , oder die kommenden Samstage seid ! RheinCandleLight Demotermine"*

Handling Punctuation

Punctutaion should only be included in the tag if they appear in the middle of sequence representing an entity.

Example: *Donnerstag[B-DATE] ,[I-DATE] 17.04.2020[I-DATE]*

Hashtags

If a hashtag is part of a named entity, it should be included in the tag.

Examples:

- #Berlin[B-ORG]
- #BW[B-ORG]

- #CH[B-ORG]
- #Hardy[B-PER] Groeneveld[I-PER]
- #Mutigmacher[B-ORG]

If a hashtag is part of a sequence combining two named entities, it should be included in the tag for the first entity.

Example: #be[B-LOC]2908[B-DATE]

Detailed Annotation Instructions for Entity Types

PERSON

Person names can consist of given name and last name, or only first / only last name. Titles (e.g. Dr.) and forms of address (Herr, Frau) should not be annotated.

Examples:

- *Scholz[B-PER]*
- *Carolin[B-PER] Matthie[I-PER]*
- *Merkel[B-PER]*
- *Anselm[B-PER] Lenz[I-PER]*
- *Karl[B-PER]*

LOCATION

Do not label derived location words (e.g. *Kreuzberger Nächte* as location.

If a a sequence contains a number of tokens representing a location entity (e.g. an address), use the B-tag for the initial token and the I-tag for all succeeding tokens belonging to the same entity, including special characters such as , - . or /. The sequence should be long enough to create a representation of an entity which can be understood without the succeeding tokens.

Examples:

- *Straße[B-LOC] des[I-LOC] 17.[I-LOC] Juni[I-LOC]*
- *Bad[B-LOC] Oldesloe[I-LOC]*
- *Elbe[B-LOC] Elster[I-LOC]*
- *Bremen[B-LOC]-Vegesack[I-LOC]*
- *Lutherstadt[B-LOC] Wittenberg[I-LOC]*
- *Landeplatz[B-LOC] Grießkirchen[I-LOC]*
- *Google[B-LOC] center[I-LOC]*
- *Stuttgarter[B-LOC] Landtag[I-LOC]*
- NOTE: *Berlin[B-LOC] Neptunbrunnen[B-LOC]*

If the sequence contains a specification of the location, include the specification using the I-tags.

Examples:

- *Königstraße[B-LOC] Richtung[I-LOC] HBF[I-LOC]*
- *Bernauer[B-LOC] Str[I-LOC] Richtung[I-LOC] Süden[I-LOC]*
- *Marktoberdorf[B-LOC]-Kaufbeuren[I-LOC]-Neugablonz[I-LOC]*
- *Bremen[B-LOC]-Vegesack[I-LOC]*
- *D-67454[B-LOC] Haßloch[I-LOC]*

ORGANIZATION

If a a sequence contains a number of tokens representing an organization entity, use the B-tag for the initial token and the I-tag for all succeeding tokens belonging to the same entity, including special characters such as , - . or /. The sequence should be long enough to create a representation of an entity which can be understood without the succeeding tokens.

Examples:

- *Bundesregierung[B-ORG]*
- *Gesundheitsamt[B-ORG]*
- *Französische[B-ORG] Polizei[I-ORG]*
- *Google[B-ORG]*
- *Bill[B-ORG] Gates[I-ORG] Foundation[I-ORG]*
- *KSV[B-ORG] LiLi[I-ORG]*
- *FC[B-ORG] St.[I-ORG] Pauli[I-ORG]*
- *Bündnis21[B-ORG] Hessen[I-ORG]*
- *Polizeipräsidium[B-ORG] Pforzheim[I-ORG]*

TIME

Time expressions can include specific times of day and time codes (usually appearing after URLs. Sequences denoting a time span / duration should be annotated with the B-tag for the initial token and I-tags for the succeeding tokens.

Examples.

- *15:00[B-TIME]*
- *17:00[B-TIME] Uhr[I-TIME]*
- *www.example.com[O] 00:30[B-TIME]*
- *Mittag[B-DATE]*

- *15:30[B-TIME] -[I-TIME] 19:30[I-TIME]*

DATE

Date expressions can be numeric values or strings. Sequences denoting a date spanshould be annotated with the B-tag for the initial token and I-tags for the succeeding tokens.

Examples:

- *2019[B-DATE]*
- *14.12.21[B-DATE]*
- *2.7.21[B-DATE]*
- *31. [B-DATE] JULI[I-DATE]*
- *01122020[B-DATE]*
- *Pfingstmontag[B-DATE]*
- *HEUTE[B-DATE]*
- *3108[B-DATE]*
- *201905[B-DATE]*
- *2019/03/05[B-DATE]*
- *28. [B-DATE]+ [I-DATE] 29.8. [I-DATE]*
- *17042021[B-DATE] –[I-DATE] 21042021[I-DATE]*

ACTION

Actions represent unique real-world events or actions. They must not necessarily be named, but be identifiable as unique object in the context in which they are mentioned. Also, the type of action should be expressed.

Examples:

- *Angriff[B-ACTION]*
- *Demo-B*
- *Autokorso[B-ACTION]*
- *Offenes[B-ACTION] Treffen[I-ACTION]*
- *Aufzug[B-ACTION] RufderTrommeln[I-ACTION]*
- *Zug[B-ACTION]*
- *Plakate[B-ACTION]*
- *Infostände[B-ACTION]*
- *Informationstouren[B-ACTION]*
- *Anti[B-ACTION]-Corona[I-ACTION]-Proteste[I-ACTION]*
- *Protestzug[B-ACTION]*
- *Spaziergänge[B-ACTION]*
- *Querdenker[B-ORG]-Demo[B-ACTION]*
- *Korso[B-ACTION] Nord[I-ACTION]*
- *Gedenken[B-ACTION]*
- *Aktionswoche[B-ACTION]*
- *Wahlkampf[B-ACTION]*
- *Live-Stream[B-ACTION]*

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.

I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.

I am aware that the violation of this regulation will lead to the failure of the thesis.

....	....
Name	Signature
....	....
Matriculation Number	Place and Date